

Appendix X: interviews to fisherman

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Methodology

Scoring

The interview notes were reviewed to identify quantitative scores applying following criteria (in this ordered):

- *C1 (score available)*: When a clear and unique self-declared score was provided, this was used as a score. In case the interviewee provided two scores (e.g.: 4 or 5), it was used the average (4.5). For those answers were alternative were provided for gear/target, create separate scoring.
- *C2 (score to be interpreted)*: Answers that were providing meaningful text but not a clear number, or that were considered worth of further attention, were scored as 999.
- *C3 (does not apply)*: Answers that were missing because the event/condition does not apply were scored as -999. When the “does not apply” was identified in more than 50% of the answers, the combination of fishing style-event was excluded from further analysis.
- *C4 (can't answer)*: Answers that were missing because people did not have a personal experience to refer but refers to an event that is relevant (i.e.: if during a storm you never go out, you might don't know about how catchability changes) were scored as -998.
- *C5 (garbled)*: Answers that were not coherent with the question were scored -997

Score	Type	Description
x	C 1	Number (x) available
999	C 2	To be interpreted
-999	C 3	Does not apply
-998	C 4	Cant answer
-997	C 5	Garbled

Interpretation

A structured coding approach was applied to those answers scored as 999 using both human expertise and artificial intelligence assistance.

- Text for unclear answers was revised to extract the main keywords (i.e.: most people say; I think it does not affect so much etc.), and to identify wether the answer was needed to support narrative or to provide quantitative information.
- The keywords were pooled based on the expected type of answer (e.g.:). Three researchers independently assigned quantitative scores to textual descriptions based on predefined LIKERT scale. These expert based ratings were averaged to create a consensus dataset, which was used to establish interpretive patterns linking descriptive terms to quantitative values.
- Answers without keywords needed for quantitative information were reviewed with Artificial Intelligence. We employed OpenAI's GPT-5 (ChatGPT, September 2025 version) as an assistant to generalize these patterns to new responses. The model was not trained on external data but was guided by the team's own scoring framework. For each new description, the model produced a suggested numerical score along with a written rationale. All outputs were reviewed and validated by the researchers to ensure consistency with the established coding scheme
- Answers without keywords that were needed to support narrative (i.e.: management) were left unscored.

Reconstruction

The questions tagged as -998 (The “I don’t know”) were revised to understand if it is possible to borrow numbers from other interviewee. In many cases people did not attempt answer questions regarding scenarios where they have never been into (i.e.: if during a storm you never go out, you might don’t know about how catchability changes).

Answers are pooled, with supervision, based on the domain of the answer: human, metier, fish, descriptive. The human domain regards aspects that depend on the specific person and can be pooled only across the same fishing style or closely related. The metier category regards fishing operations and can be pooled for similar gears and seasons. The fish category regards ecological aspects of the fish and can be grouped by fish type and season.

Following tasks: revise the narrative

- Multi gear/multiarea: some fisherman switch gear during the season and/or because of the event, and the alternative gear has different implications in terms of target species and risks. This thing of the switching gear/species is particularly relevant for sport fisherman and it can be coupled in case study 2. I am working on the DAG.
- In terms of species we might limit to the most important and shared, which are perch, pike, herring, whitefish, sea trout, salmon. Some people mention other species. Like zander, vendace, eel, bream.
- Interactions: the only relevant interactions mentioned were wind with rain (summer) or ice (winter). It was not mentioned often a quantitative answer for this interaction but in general people says that became more dangerous and less likely to go fishing.
- The wind direction: they mention it so much that it might be worth. I am trying to think about this.
- Lag effect. The main lag effect mentioned is what happen after a storm. Depending on the wind direction the catchability can improve or not the days after the event.
- Create a price list if possible, and consider personal time as a variable: the prices (so far) has been quite detailed reported. This can help us in doing better. When you get the numbers, see if possible to talk with edconomists. This can also maybe include the cost per work per hour. Is this too much? Or we can ghather this as a extra variable (personal time?). Question on gear cleaning always unclear. On the extra fuel: in some case is just the extra, in other is the total for the long travel. Double check and harminise this o Extra consequences: They also keep mentioning that wind can blow things away, this was not really initially considered. Many mention the positive effect of wind when it blows from certain directions. This is kind of a long term effect

Results

Scoring - overview

The questionnaires originated more than 500 scores. About half of the responses (266) were including a clear and unequivocal quantitative score. A large number (123) were not considered relevant by interviewee. Answers that were considered relevant but did not provide a clear score were 150, among which the large majority (105) contained enough information to attempt an interpretation.

Score	Frequency
x	266
-999	123
999	105
-998	35
-997	5

Scoring - does not apply (-999)

For each interviewee and event we computed the proportion of the answers scored as -999. These events with more than 50% of answers scored as -999 were excluded. The combination of events/styles excluded are the reported in the table. The image reports the proportion of answers indicating that the event is relevant.

desc	style
Heatwave in the winter season	coastal
Heatwave in the winter season	coastal
Severe icing (in the winter season)	coastal
Severe icing (in the winter season)	subsistence
Severe icing (in the winter season)	coastal
Algal bloom (in the summer season)	coastal

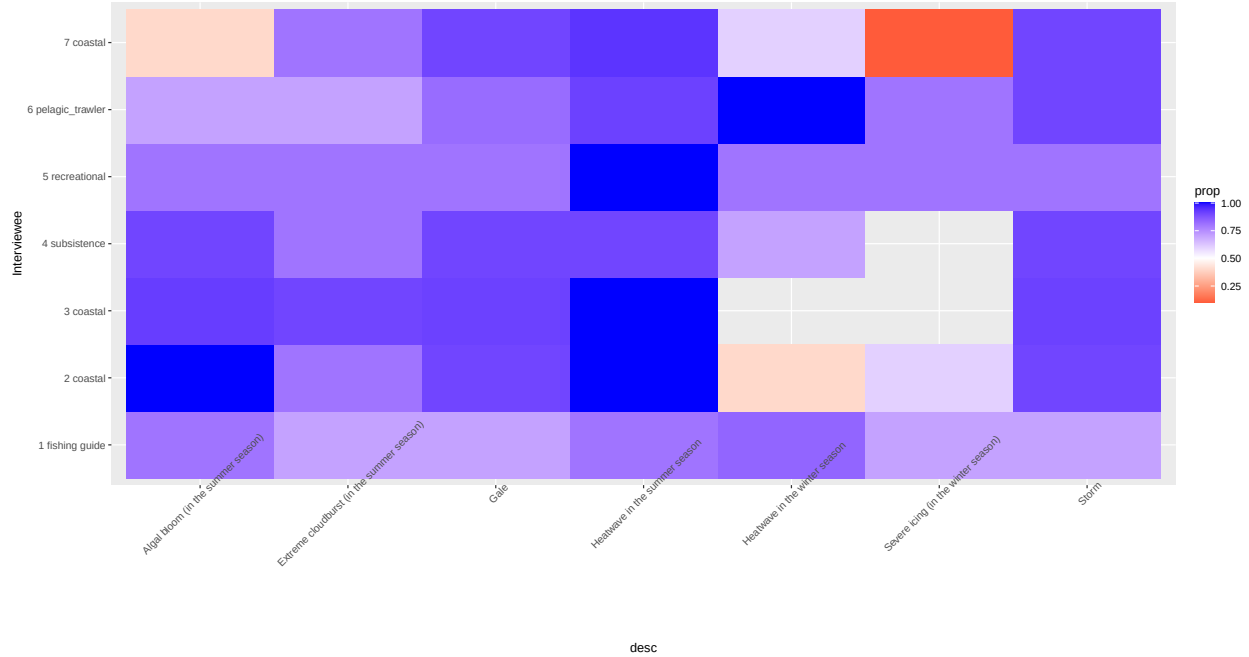


Figure 1: Proportion of responses that were considered relevant by interviewee and event

Scoring - Garbled (-997)

These answers were clearly not matching the question. We attempted a reconstruction as described in next phases. Most of these relates to the catch composition: we were interested in the behavior of the fish, not of the fisherman. Anyway we can use this information to support other aspects (e.g.: how easy is to avoid an event).

style	description	text
fishing guide	Do you catch a different composition of species during such extreme weather events?	: 5 sea trout on open water, instead of perch and pike under the ice
fishing guide	Do you catch a different composition of species during such extreme weather events?	: 5 redirect the fishing down below the thermocline
fishing guide	Do you catch a different composition of species during such extreme weather events?	More about changing location (deeper), 5 choose other species (we fish pike & perch, sea trout)
fishing guide	Do you catch a different composition of species during such extreme weather events?	: 2 sometimes; if it blows too hard and we are supposed to fish trout, then sometimes we have to go in and fish perch & pike instead
fishing guide	How much do you think the answers to these questions (questions 2-7) vary between fishers in the group you represent? [1-5; from 1 – not at all, to 5 – very much]	5, since I am chairman I have the mandate to represent the others

recreational	Do you experience that the catchability of the fish changes due to such extreme weather events?	If there's no ice, you can't catch them with an ice rod, and you don't put in a boat just for a short ice-free spell.
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Interpretation - Expert scoring

Expert scoring of keywords produced the following results, which were used to assign scores to 87 answers:

question	coding	answer	AG	ENA	JO	mean	sd
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	hard	2	2	2	2	0.0
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	easier	3	4	4	4	0.6
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	to some extent	3	4	3	3	0.6
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	no	1	1	1	1	0.0
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	not difficult	4	3	5	4	1.0
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	it is possible	3	4	4	4	0.6
is it possible [to avoid something]	1-5; from 1 – very difficult, to 5 – very easy	probably not entirely impossible	2	2	2	2	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	you can't [do something]	1	1	1	1	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	yes but [in a different way]	3	3	3	3	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	it has a big negative impact	1	1	2	1	0.6
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	still possible [to do something]	3	3	3	3	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	most do not [do something]	no info	2	2	2	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	impossible [to do something]	1	1	1	1	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	nothing can be done	1	1	1	1	0.0

is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	we can do little	2 or 4, depending on Swedish wording (we can do little vs we can do a little)	2	3	2	0.7
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	I would try to	no info	3	4	4	0.7
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	we wait	no info	1	??	1	NA
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	sometimes	3	3	3	3	0.0
is it possible to do [something]	1-5; from 1 – very unlikely, to 5 – very likely	i'll do it anyway	no info	4	4	4	0.0
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	very similar	1	1	1	1	0.0
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	small difference	2	2	2	2	0.0
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	it may be different	4	4	3	4	0.6
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	it has an effect	no info on direction	3	4	4	0.7
is it [something] different when [event] happens	1-5; from 1 – very similar, to 5 – very different	it affects	no info on direction	3	4	4	0.7
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it is harder to do	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	there is a marginal negative effect	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it is more difficult	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it get worse	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	there is a little negative impact	2	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	not so big impact	3	2	3	3	0.6

does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	a little worrisome	no info	2	2	2	0.0
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	it certainly has a negative effect	2	1	1	1	0.6
does [event] makes [something] worse or better	1-5; from 1 – much worse, 3 – no difference, to 5 – much better	is good	4	4	4	4	0.0
how much is [something] important	1-5; from 1 – not important at all, to 5 – very important	fairly important	3	3	3	3	0.0
how much is [something] important	1-5; from 1 – not important at all, to 5 – very important	not a strong link	2	2	2	2	0.0
how much is [something] important	1-5; from 1 – not important at all, to 5 – very important	it has a real significance	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it has great effect	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it is big effect	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it may have an effect	3	2	3	3	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it has big impact	5	4	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	Quite small	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	not so much	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	not really	1	1	1	1	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it does	3	2	3	3	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	quite a lot	4	4	4	4	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	some	2	3	3	3	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	some, but less than other	2	2	2	2	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	greatly	4	5	4	4	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	most do not	2	2	2	2	0.0

does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	no	1	1	1	1	0.0
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	improve	4	3	4	4	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	a lot	4	5	5	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	it's strong	5	5	4	5	0.6
does [something] has an effect	1-5; from 1 – not at all, to 5 – very much	pretty much	4	4	4	4	0.0
How much concern for your personal safety	0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening	big concern	no info bc difference between 1 & 2 is type of concern, not level alone	2	2	2	0.0
How much concern for your personal safety	0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening	some risk	no info bc difference between 1 & 2 is type of concern, not level alone	1	1	1	0.0
How much concern for your personal safety	0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening	worrisome	no info bc difference between 1 & 2 is type of concern, not level alone	1	1	1	0.0
does [something] allows you to change location	0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible	sometimes, but not always	1	1	1	1	0.0
does [something] allows you to change location	0-2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible	it allows a lot of freedom	2	2	2	2	0.0

Interpretation - AI scoring

This option was applied to a limited number of responses.

Q1

Question: How much do you think they vary among the group of people you represent? Possible answers: 1–5; from 1 – not at all, to 5 – very much

Observed Answer: it is so personal, some do not care at all, they just drink beer all day; we discuss things like this all the time, most are probably quite stressed by such situations

Score: 4

Why: The reply indicates clear heterogeneity — a minority who “do not care at all” vs a majority who are “quite stressed.” That is meaningful variation but phrased so that most are similar (stressed) with a minority different. In your training examples, minority exceptions vs majority agreement tended to map to a high-but-not-maximum value. Confidence: medium–high.

Q2

Question: How much do you think they vary among the group of people you represent? Possible answers: 1–5; from 1 – not at all, to 5 – very much

Observed Answer: There are some who have rights but don’t practice at all, and then some who practice like I do, just now and then. Then there are those who have leased out their permits, and in those cases, there is somebody who practices quite intensively

Score: 5

Why: The answer describes the full spectrum (no practice → occasional → very intensive). That explicit broad distribution matches your examples that consistently received top-end variability scores. Confidence: high. Q2 *Question:* How likely is it that you explore new techniques or strategies? Possible answers: [1–5; from 1 – very unlikely, to 5 – very likely] *Observed Answer:* Would probably try to find solutions; activity would certainly change, and then people would follow that change; there is probably some variation among people depending on what one is specialized in *Score:* 4 *Why:* The speaker says they “would probably try” and expects change and follow-up by others — language that in your dataset equates to a positive likelihood but not the absolute “definitely” (which you typically scored a 5). The note about variation reduces absolute certainty slightly. Confidence: medium–high.

Q3

Question: If your income from activity A decreases, can you complement or replace it? Possible answers: [1–5; from 1 – very unlikely, to 5 – very likely]

Observed Answer: It would be very hard to replace activity A it with other income. Most have other jobs in winter, but A is such a significant part of income

Score: 1

Why: The phrase “very hard to replace” and “A is such a significant part” matches wording in your dataset that mapped to the low end (very unlikely) — even though there are seasonal other jobs, the respondent explicitly indicates replacement is essentially not feasible. Confidence: high.

Q4

Question: How easy is it to avoid an extreme weather event by going to another location?

Possible answers: [1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid] *Observed Answer:* I Stop my activity *Score:* 1 *Why:* The respondent says they stop the activity instead of relocating to avoid the event — that implies relocation is not used/feasible. In your training set, answers that indicate “we don’t/stop” were scored at the difficult end (1). Confidence: medium — short reply is slightly ambiguous about whether “stop” means temporarily pause or that relocation is impossible, but both interpretations point to very low ease of avoiding by moving.

Q5 Question: If the following type of extreme event occurs, how likely is it that you go out and do the activity? Possible answers: [1–5; from 1 – very unlikely, to 5 – very likely] *Observed Answer:* we wait because the product can spoil *Score:* 1 *Why:* “We wait” because of spoilage risk indicates they do not go out during the event — i.e., very unlikely. In your dataset, similar phrasing (“we wait / won’t go because...”) mapped to the lowest likelihood. Confidence: high.

Interpretation - left unscored

These were left unscored either because are mainly needed to create narrative or because their implementation would require further revision of the network. For example there is a lot of mentions to wind direction, which is not yet implemented.

style	description	text
coastal	Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.	Generally, we have quite a lot of mobility in terms of management; fixed gear locations are accessible fishing spots. Only for salmon fishing the license is tied to a specific boat, so you can’t switch to a larger boat. For net fishing it doesn’t matter. (Example given: switching to a larger and safer boat than the one tied to the salmon license in order to haul the salmon trap during gale, but then getting a penalty on the license).
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Yes – after a gale from the right direction, herring can blow in, salmon may swim more too (E, NE in summer). =4. From W & NW it gets worse instead.
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Yes – after a gale from the right direction, herring can blow in, salmon may swim more too (E, NE in summer). =4. From W & NW it gets worse instead.
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Same, just stronger wind.
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Same, just stronger wind.
coastal	How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]	Could damage or destroy gear/boat, or cause minor damage
coastal	How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]	Could damage or destroy gear/boat, or cause minor damage
coastal	Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?	Yes, can move around for herring, whitefish, perch. Only salmon gear is fixed. Our operations use active gear, so we can relocate, but only short distances – catch must be landed quickly. (Older generations even traveled to Trosa/Grisslehamn.)

subsistence	Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.	Eel fishing ban; voluntary fishing stop Mar 15–May 15 in some spawning bays. We must restore predatory fish stocks. We think fish shelters (we’ve built 10 over 10 years) have limited effect at such low stock levels.
recreational	Do you catch a different composition of species during such extreme weather events?	Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season
recreational	Do you catch a different composition of species during such extreme weather events?	Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season
recreational	How much does your “community” benefit from your fishing? Explain!	Some drive among those eager to show off successful fishing. But being part of that community doesn’t really require fishing.
coastal	Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.	There isn’t much else to fish for; we catch other fish species, but in much smaller quantities, such as pike and pikeperch; salmon is subject to quotas; herring too, but we don’t fill the coastal quota, and haven’t done so for several years.

Reconstruction (-998)

For these questions we try to inspect closely related answers to see if we can borrow the scoring. These numbers are stored and then used to train the BN (so are not reported in section XXX)

- “Personal” feelings

Personal feelings answers deal with personal decisions taken in face of an event. This is personal in first place and so any “borrowing” should be done with caution. We can inspect closely related fishing styles. We have only two records that needs to borrow from other in the Human category.

style	description	desc	value
coastal	How much concern for your personal safety do these extreme weather events bring?	Storm	-998
recreational	How easy is it to avoid an extreme weather event by going to another fishing location?	Severe icing (in the winter season)	-998

The coastal fisherman category is represented by three interviewee. We do not need to reconstruct anything at this stage because the BN can be informed by the other respondents. For the recreational fisherman we can try to look for the same answer in the fishing guide. Unfortunately this reflecting too much the commercial purpose of the fishing guide and cannot be applied to the recreational.

description	text	value
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How easy is it to avoid an extreme weather event by going to another fishing location?	Easy for me to switch but ask what the customers want (e.g., Americans who are booked for ice fishing). Large differences from the normal are a problem, with bookings we do not know what we shall book (e.g., closed Nyköpingsån fishing due to heatwave). The customers also have to be flexible. 2 for those that are not dangerous; 1 for what is dangerous (cannot conduct activities)	1
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- metier characteristics

Metier characteristics answers deal with technical aspect of the fishing gear or the gear/fish interaction in face of an event. A similar gear can be used by different categories (e.g.: nets, rods). Also, some gears (salmon traps) are very specific and only some people do mention. We have a handful of missing answers (total = 7).

style	description	desc	value
recreational	Do you experience that the catchability of the fish changes due to such extreme weather events?	Heatwave in the winter season	-997
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Severe icing (in the winter season)	-998
coastal	Do you experience that the catchability of the fish changes due to such extreme weather events?	Algal bloom (in the summer season)	-998
recreational	Do you experience that the catchability of the fish changes due to such extreme weather events?	Algal bloom (in the summer season)	-998
fishing guide	Do you experience that the catchability of the fish changes due to such extreme weather events?	Extreme cloudburst (in the summer season)	-998
fishing guide	Do you experience that the catchability of the fish changes due to such extreme weather events?	Storm	-998
recreational	Do you experience that the catchability of the fish changes due to such extreme weather events?	Storm	-998

The coastal fisherman category is represented by three interviewee. We do not need to reconstruct anything at this stage because the BN can be informed by the other respondents. For the recreational and fishing guide, they can inform each other for the effects of winter heatwaves and cloudburst on catchability.

style	gear	description	desc	text	value
fishing guide	ice fishing	Do you experience that the catchability of the fish changes due to such extreme weather events?	Heatwave in the winter season	1 ice fishing, there is no catchability if there are no fish; however, the fish like when it gets warmer, so if it's spin fishing then 5	1
fishing guide	rod	Do you experience that the catchability of the fish changes due to such extreme weather events?	Heatwave in the winter season	1 ice fishing, there is no catchability if there are no fish; however, the fish like when it gets warmer, so if it's spin fishing then 5	5

recreational	NA	Do you experience that the catchability of the fish changes due to such extreme weather events?	Extreme cloudburst (in the summer season)	No.	3
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- Fish related

The category fish related inform on ecological or physiological characteristics of the fish, which are expected to not be dependent on gear or people. We can try to borrow from other answers dealing with similar species and environment. We have a variety of missing answers falling in this category. The main problems are with species composition ($n=X$) and fish condition ($n=Y$). Since this affects the fish, we can inform the missing answers depending on people that attempted a score.

style	description	desc	value
fishing guide	Do you catch a different composition of species during such extreme weather events?	Heatwave in the winter season	-997
subsistence	Do you catch a different composition of species during such extreme weather events?	Heatwave in the winter season	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Heatwave in the summer season	-997
subsistence	Do you catch a different composition of species during such extreme weather events?	Heatwave in the summer season	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Severe icing (in the winter season)	-998
coastal	Do you catch a different composition of species during such extreme weather events?	Severe icing (in the winter season)	-998
recreational	Do you catch a different composition of species during such extreme weather events?	Severe icing (in the winter season)	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Algal bloom (in the summer season)	-997
coastal	Do you catch a different composition of species during such extreme weather events?	Algal bloom (in the summer season)	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Extreme cloudburst (in the summer season)	-998
subsistence	Do you catch a different composition of species during such extreme weather events?	Extreme cloudburst (in the summer season)	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Gale	-997
subsistence	Do you catch a different composition of species during such extreme weather events?	Gale	-998
coastal	Do you catch a different composition of species during such extreme weather events?	Gale	-998
fishing guide	Do you catch a different composition of species during such extreme weather events?	Storm	-998
coastal	Do you catch a different composition of species during such extreme weather events?	Storm	-998

subsistence	Do you catch a different composition of species during such extreme weather events?	Storm	-998
coastal	Do you catch a different composition of species during such extreme weather events?	Storm	-998
recreational	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Severe icing (in the winter season)	-998
subsistence	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Algal bloom (in the summer season)	-998
coastal	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Extreme cloudburst (in the summer season)	-998
subsistence	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Extreme cloudburst (in the summer season)	-998
coastal	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Gale	-998
subsistence	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Gale	-998
coastal	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Storm	-998
subsistence	Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?	Storm	-998

By inspecting the distribution of answers regarding species captured it emerge consistency, with some more variability observed in summer heatwave (expected). We observe one outlier for algal blooms. This is to be revised.

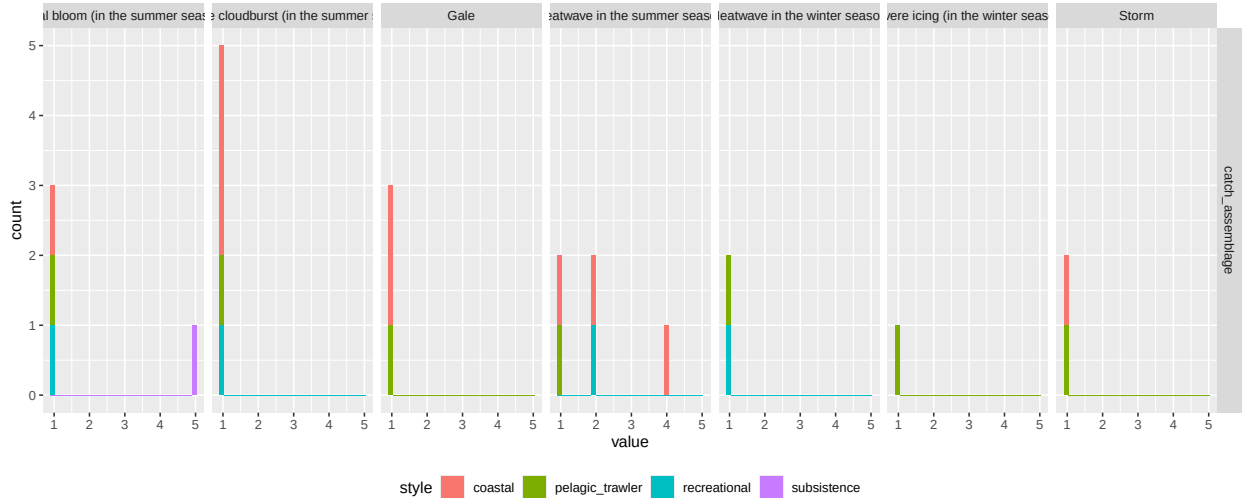


Figure 2: Are the fish in the catch of worse condition or quality [...]?- From 1 much worse to 3 same to 5 much better

By inspecting the distribution of answers regarding fish condition, it emerge a pattern that regards pelagic

trawler. This does not apply to other people (it is damage to fish due to rolling in big waves). Other responses can be used.

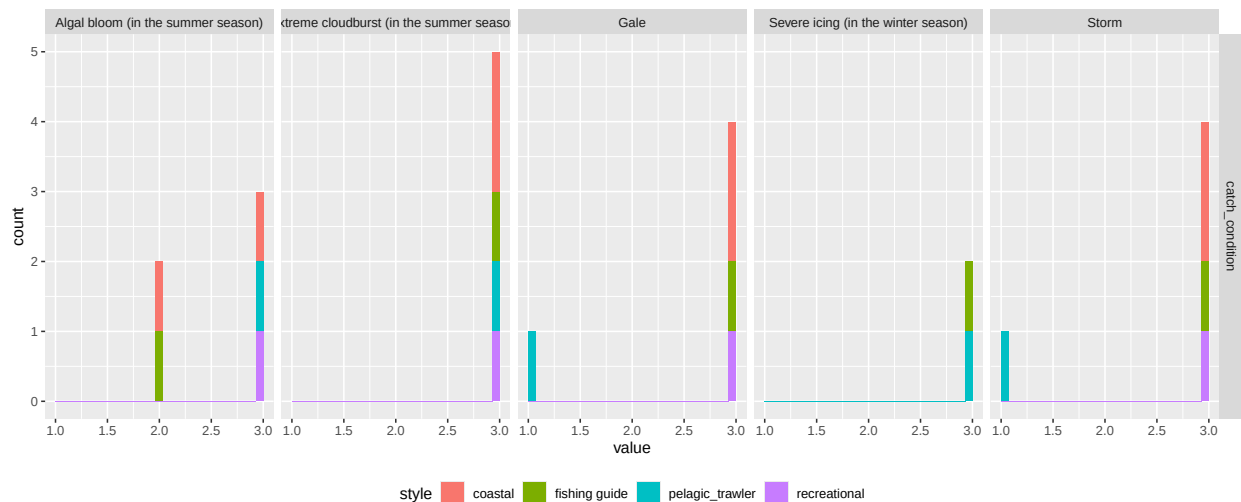


Figure 3: Distribution of answers to the question: Do you catch a different composition of species during such extreme weather events?- From 1 very similar to 5 very different

- Other

What remains are questions regarding uncertainty, long term effects, and change in fishing practices. Long term effects and change in fishing practices are to be used to support narrative and we do not really need them at this stage. It is more important to have some information on uncertainty-related questions.

style	description	desc	value
coastal	Do this extreme weather events have long-term effects on how you fish or catch composition	Severe icing (in the winter season)	-998
coastal	Do this extreme weather events have long-term effects on how you fish or catch composition	Algal bloom (in the summer season)	-998
coastal	Do this extreme weather events have long-term effects on how you fish or catch composition	Extreme cloudburst (in the summer season)	-998
fishing guide	How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? [1–5; from 1 – not at all, to 5 – very much]	NA	-997
recreational	Do you change anything in how you normally fish when such an extreme weather event occurs?	Severe icing (in the winter season)	-998
coastal	Do you change anything in how you normally fish when such an extreme weather event occurs?	Storm	-998
subsistence	How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent	NA	-998

Revision of the narrative

We have now all the information and it is time to include it in the network. For this purpose we might have to adapt the last version of the network. Something will be excluded and something added. I will try to follow these principles (Korb and Nicholson, 2011):

- Node values are mutually exclusive: if two things can happen at the same time, they have to have a node each one (e.g.: the events should be separate nodes)
- Node values are exhaustive: contains all the possible options (so add “No” or “Other” when necessary)
- Do not include complexity that cannot be modeled: if some combination of events/consequences results not important across most of the interviews, then exclude nodes or links

Treatment of multiple options

I identified all the mentions to alternative options.

The gear/area can be used to:

- Avoid the event (switching from ice fishing to spinning; casting the nets deeper etc..). There are consequences in terms of cost and, eventually, target species
- Set up scenarios: coastal fisherman seems to plan different activities during the season. Especially for what regard salmon. Maybe we can be explicit for the consequences of salmon fishing compared to net fishing

The options regarding fish can be used to define the probability to find a species given gear, area and event (only for summer heatwave); as well as to inform the condition (also only for heatwave)

Interactions

Very few interactions were mentioned. Considering that each event will be a node, there is no problem in implementing interactions.

style	description	desc	text
fishing guide	If the following type of extreme event occurs, how likely is it that you go out and fish?	Are there any of these in combination that affect how likely it is that you go out and fish?	A lot is about safety, combinations become dangerous, e.g., strong icing and strong winds, I avoid this as much as possible; would also get fewer requests, since fishing should be enjoyable
fishing guide	How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]	Extreme cloudburst (in the summer season)	don't know, more if the boat sinks; is that a 3 then, don't know, depends on how big the damage is; in combination with storm then it becomes 3. Usually, you get away from a cloudburst. There is a risk; maybe 2
coastal	If the following type of extreme event occurs, how likely is it that you go out and fish?	Are there any of these in combination that affect how likely it is that you go out and fish?	Not really; it's the wind that is decisive for us

coastal	If the following type of extreme event occurs, how likely is it that you go out and fish?	Are there any of these in combination that affect how likely it is that you go out and fish?	Wind has the greatest impact here. Wind direction matters too (this August mostly W & NW winds, cooling the water – no point fishing perch then, they just stop moving).
subsistence	If the following type of extreme event occurs, how likely is it that you go out and fish?	Are there any of these in combination that affect how likely it is that you go out and fish?	Strong winds and high water levels.
recreational	If the following type of extreme event occurs, how likely is it that you go out and fish?	Are there any of these in combination that affect how likely it is that you go out and fish?	Combination of gale/storm and heavy rain – Even more unpleasant, a negative synergy effect
pelagic_trawle	If the following type of extreme event occurs, how likely is it that you go out and fish?	Are there any of these in combination that affect how likely it is that you go out and fish?	Icing & high winds, then it's 1
pelagic_trawle	How much concern for your personal safety do these extreme weather events bring?	Heatwave in the winter season	1 if the heat causes high winds
pelagic_trawle	How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]	Heatwave in the winter season	3, [heat wave in winter, often accompanied by storms]
coastal	If the following type of extreme event occurs, how likely is it that you go out and fish?	Are there any of these in combination that affect how likely it is that you go out and fish?	storms and weather like that

Probabilities

Most of the mentions to probability are regarding the wind. Still thinking about this.

List still to be indexed:

I2: (storm mostly damage) to gear. It very rarely happens that we lose gear completely, but a small risk exists

I3: Wind direction matters too (this August mostly W & NW winds, cooling the water – no point fishing perch then, they just stop moving)

I3: storm and gale Could damage or destroy gear/boat, or cause minor damage

I6: Heatwave in the winter season: It can go both better and worse, but it has an impact; for the same weather phenomenon, it can be both better and worse; weather changes affect catchability, as do moon phases and the time of year.

I7: Gale: depends on the wind direction, if it's N then it's from the outside so then it's difficult, if it's S or SW, in gale force winds I don't fish anyway because there's a strong current in the lake so you get crap in

your nets, but I do go out in gale force winds sometimes, depends on the wind direction

Lags

The lags can perhaps be implemented externally. I will skip this part at this stage.

I1; cloudburst: fish react to pressure changes; if there is low pressure during a downpour, then they stop biting, but if it rises after the downpour then it can be good.

I3 (gale): after a gale from the right direction, herring can blow in, salmon may swim more too (E, NE in summer). =4. From W & NW it gets worse instead

I5: the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal

I5: sea trout fishing is big – if warming lasts longer, the season gets shorter until it cools again

I4: Other factor is wind direction, NW is bad, then the fish disappear. Big impact. This spring it locked in at N & NW with near gale winds for several weeks

I6: force, and waits 1-2 days until it calms down; when the waves are too high, the trawl does not fish as well

Identifying which links matters to the DAG

This is documentation on the decisions taken to draw links between events and outcomes.

We inspect the responses per event on each single potential response. We decide which to retain based on importance and sometimes agreement. As we did for the answers to be interpreted, we can identify (i) personal aspects; (ii) métier characteristics and (iii) fish response. Each answer regarding personal answer cannot be questioned and should be included as much as possible. What regard métier and fish can be revised more carefully (e.g.: the effect of algal blooms).

go_out

Going out or not is a personal decision. This link is the most important probably and we keep everything.

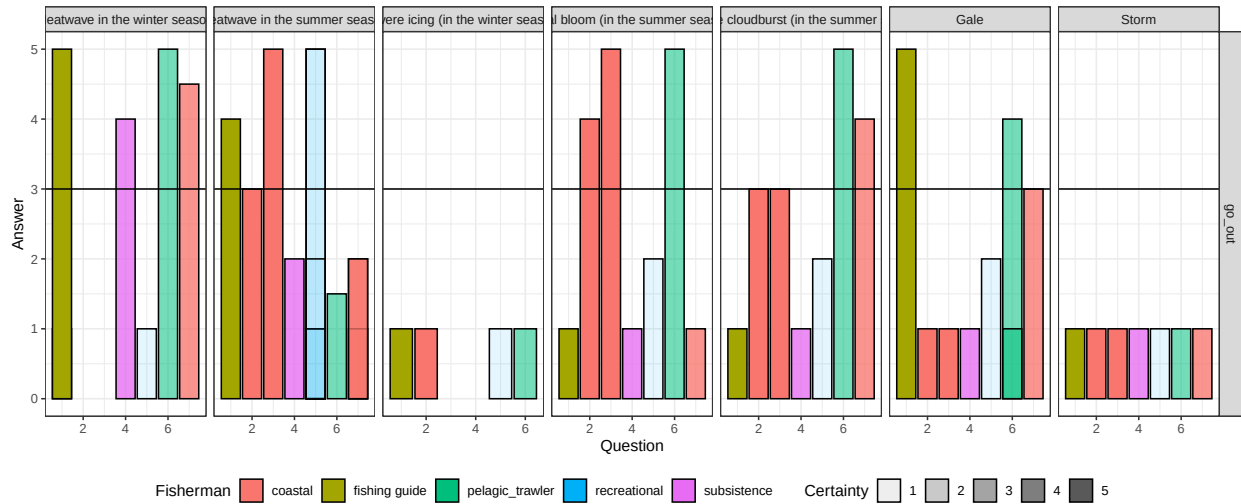


Figure 4: Distribution of answers to the question: go_out

avoidance

For each event there is at least somebody that say he can avoid it. This typically imply change fishing location, sometime /especially recreational) to change gear.

personal_safety

Personal safety NOT apply to algal bloom and to summer heatwaves. Winter heatwaves is associated with ice fishing. The mention that pelagic trawler do is not correct (considers an interaction with storms).

damage

The gear damage that pelagic trawler mention for winter heatwave depend on the storminess, not on the heat itself. For heatwave, int N 2 mention extra work to clean, not real damages. Same for algal bloom ant int 6. At this stage we are not considering "extra work/maintenance" as part of the damages. Why: only a few mention this here. Some other mention this in extra or here and there. Revise it.

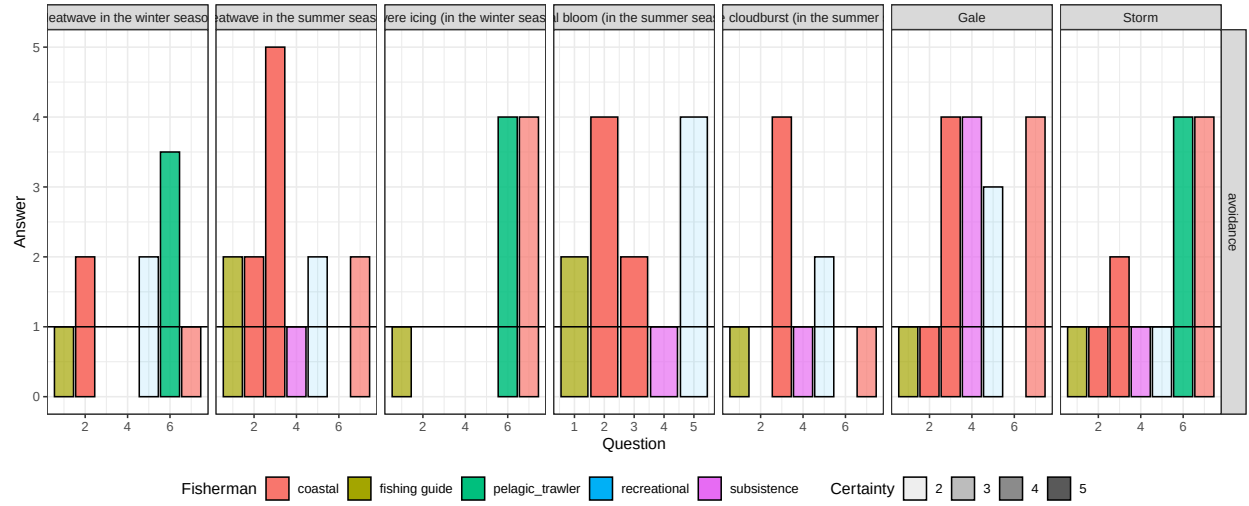


Figure 5: Distribution of answers to the question: avoidance. Answ: 1 is very difficult, 5 is easy.

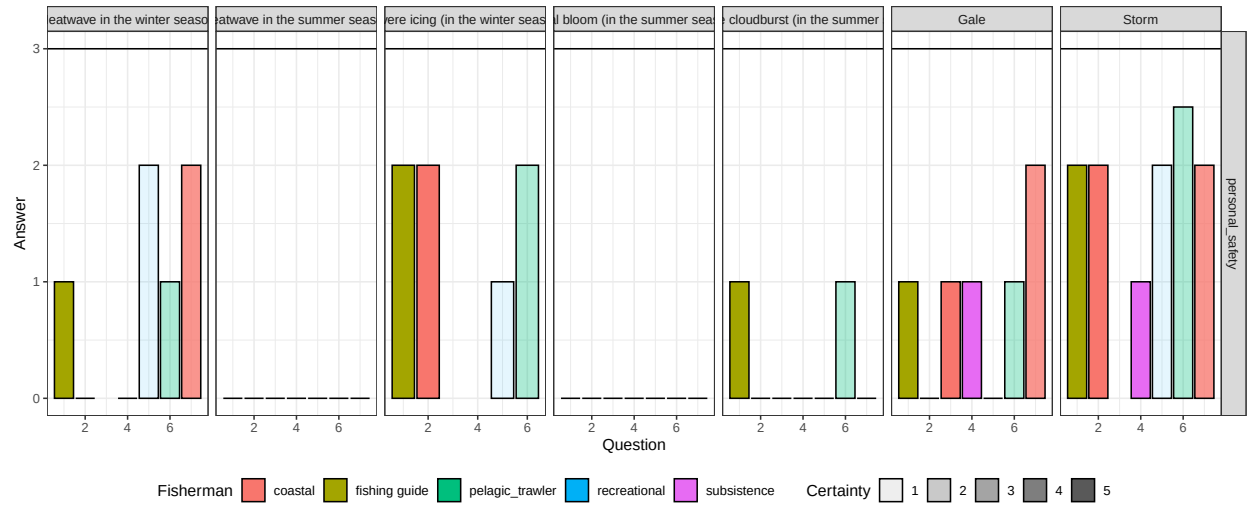


Figure 6: Distribution of answers to the question: personal_safety

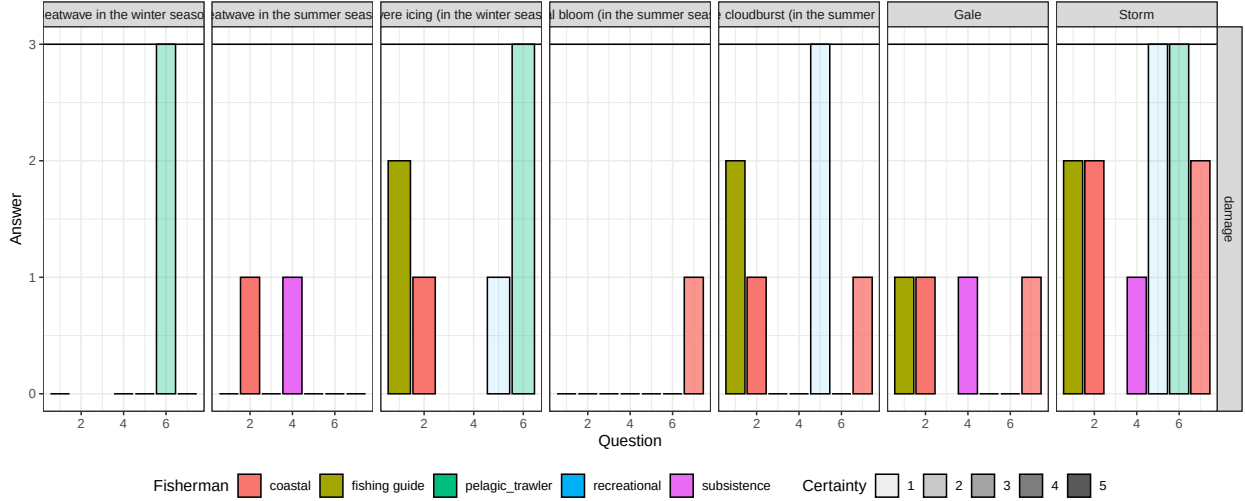


Figure 7: Distribution of answers to the question: damage

Catchability

Catchability is associated with all the events except for cloudburst. Note: winter heatwave change for fishing guide is associated with rod fishing

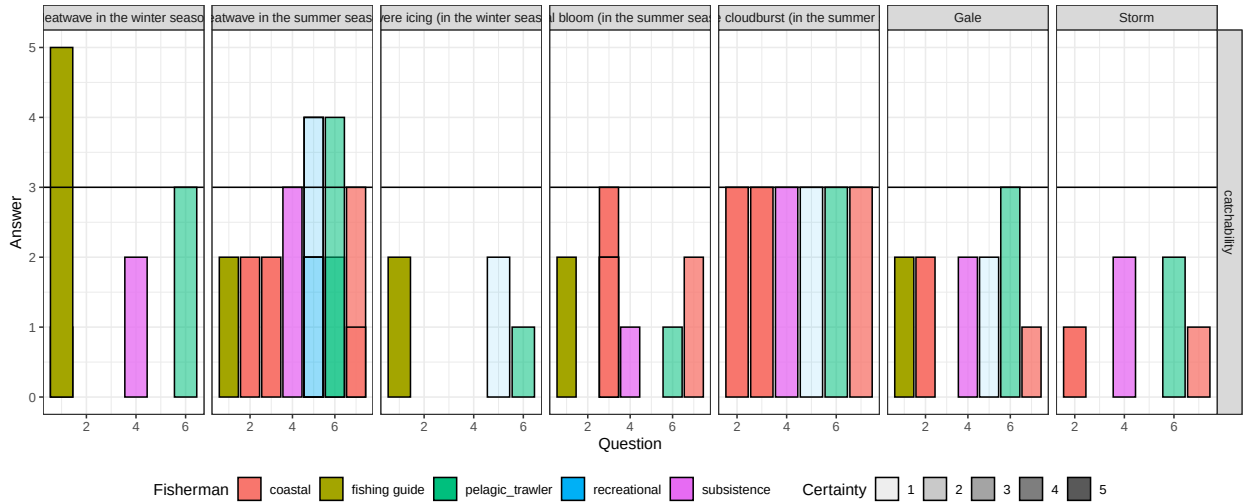


Figure 8: Distribution of answers to the question: catchability

catch_condition

For condition we ignore winter heatwave, because that is the effect of storminess. We keep storms and gales because it makes sense (keeping the catch onboard in a storm may cause the fish the get bad). We also exclude algal blooms because it matters only for two person, and one says that it is only hypothetical.

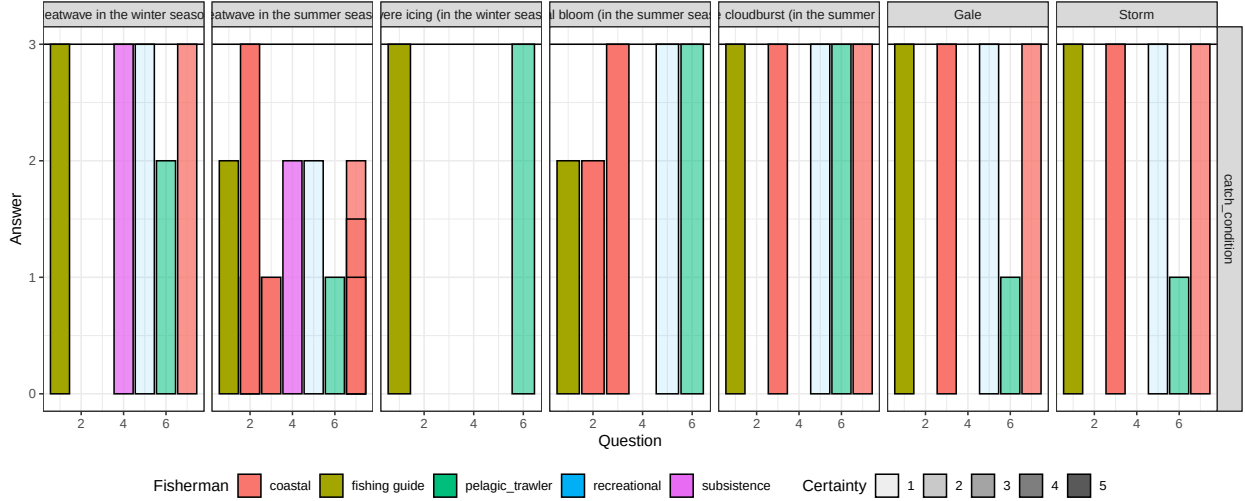


Figure 9: Distribution of answers to the question: `catch_condition`

cope condition

For cope condition we do not want to consider those answers that says “go fishing elsewhere”. Because we account for this in another manner. We only keep the link with summer heatwave because it contains things such as ice machines, short hauls etc. So we revise the network and make the

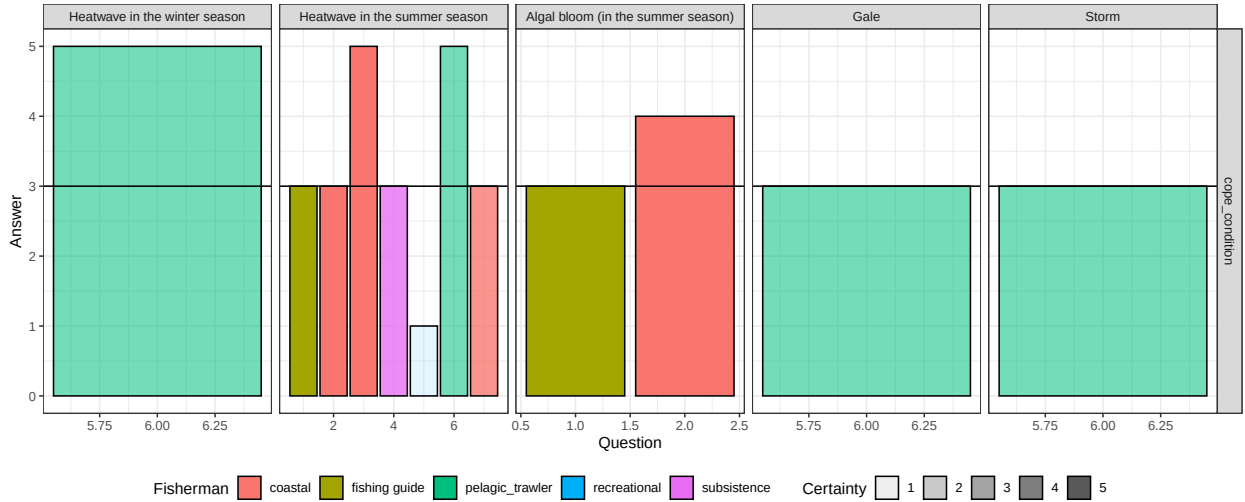


Figure 10: Distribution of answers to the question: `cope_condition`

catch_assemblage

For condition we ignore all but summer heatwave. There are meaningful answer about it: a coastal fisherman get more breams, another mention change in distribution between herrinh and perch. We exclude algal blooms because it matters only for one person, and it says that it is only hypothetical.

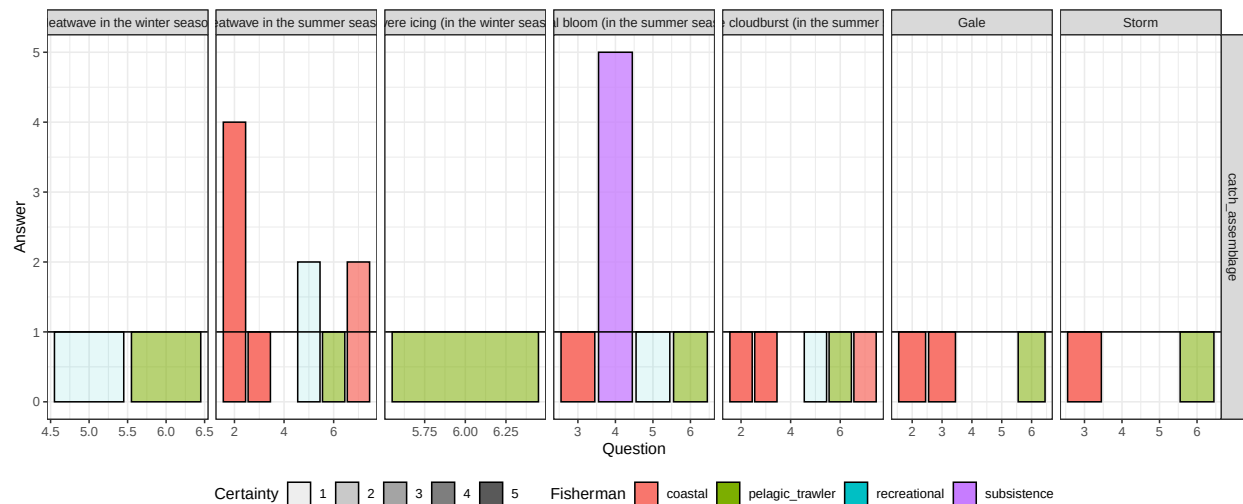


Figure 11: Distribution of answers to the question: catch_assemblage

pending decisions

- is extra work part of the damages?

Interviews - integral text (English)

Legend:

Score	Type	Description
x	C 1	Number (x) available
999	C 2	To be interpreted
-999	C 3	Does not apply
-998	C 4	Cant answer
-997	C 5	Garbled

Interviewee: I 1

1. name: peter berggren. category: fishing guide id1 is a fishing guide focused on the waters around stockholm, both in the archipelago and in lakes. he guides mainly for trout, pike, and perch, and makes a living from his business. on average, they guide about 100 days per year. the guiding business is seasonal and follows the fish's behaviour, activity, and migrations. they fish in lakes, streams and coastal areas. they can use boats in coastal areas and lakes. it is more common in spring and summer. in winter they guide ice fishing. if there is no ice they can switch to spinning but this is unpredictable. the most fished species are sea trout, salmon, pike, and perch. they primarily practice catch and release fishing. because fishing guide is mostly a touristic activity, it is more active in the weekend, and their flexibility is linked to customers flexibility. how they adapt: fishing guide adapts all the time according to the conditions and the clients need. they are flexible for place, species and gear, but its ability to adapt to conditions depends on client's flexibility. he notices a decline in demand during extreme hot weather, and for security reason he cannot guide if he feels danger. in winter he can replace ice fishing with spinning but cannot book this in advance. in summer he can choose between different options depending on the conditions. he can change fish location and target (and accordingly gear) in heatwaves, algal bloom and stormy conditions. in hw he can move the trips to mornings and evenings. in hw and ab he moves from shallow waters for pike to below the thermocline for sea trout; in stormy conditions he goes the other way around.

Q 2 - go_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1-5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Very different, depends on what is booked. Sometimes ice fishing is booked and then we have to cancel all bookings, sometimes open water fishing is possible. Lose ice fishing customers and gain spin fishing ones; 1 ice fishing (spin fishing customers can't be booked in advance in winter, you can't count on a heatwave in winter)	NA	M
b	Heatwave in the summer season	4. Some fishing is closed if it gets too hot, e.g., river fishing. It's difficult to fish if it gets extremely hot, but I try to adapt; so far I haven't canceled coastal fishing, I move it to mornings & evenings. But, for example, Nyköpingsån was closed because the water was too warm. Also depends on how hot it is. It happened in 2018, depends on what happens with our fish	4	O

c	Severe icing (in the winter season)	1. Haven't experienced severe icing, only seen it on YouTube; don't know if it happens on sportfishing boats; it has to be safe, not dangerous; if it's icing like on the Great Lakes, then the boat breaks	1	O
d	Algal bloom (in the summer season)	1. Usually happens offshore; can't handle it if I have a trip, e.g., from Sandhamn, but then we can go inland when there's algal bloom	1	O
e	Extreme cloudburst (in the summer season)	1, dangerous	1	O
f	Gale	5Gale is normal (sometimes we cancel), from strong gale and upwards we stay home (18 19 m/s average and 25 m/s in gusts is not possible)	5	O
g	Storm	1	1	O
h	Are there any of these in combination that affect how likely it is that you go out and fish?	A lot is about safety, combinations become dangerous, e.g., strong icing and strong winds, I avoid this as much as possible; would also get fewer requests, since fishing should be enjoyable	NA	NA

Multiple options identified

	Event	gear	Score (given)
a	Heatwave in the winter season	ice fishing	1
a	Heatwave in the winter season	rod	5

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better]

	Event	Answer	Score	Method
a	Heatwave in the winter season	1 ice fishing, there is no catchability if there are no fish; however, the fish like when it gets warmer, so if it's spin fishing then 5	NA	M
b	Heatwave in the summer season	2	2	O
c	Severe icing (in the winter season)	No, can't conduct fishing; if it is possible to fish then it becomes 2.	2	O
d	Algal bloom (in the summer season)	2, the fish move from that area; also becomes difficult fishing (fished during an algal bloom this summer, had to go all the way into the bay because where we intended to fish we only got weeds on the hook).	2	O
e	Extreme cloudburst (in the summer season)	Don't go out in cloudbursts; fish react to pressure changes; if there is low pressure during a downpour, then they stop biting, but if it rises after the downpour then it can be good; so I can't say	-998	NA

f	Gale	: 2, a gale doesn't matter for the fishing itself, other than that the gear is hard to use	2	O
g	Storm	Don't go out in a storm	-998	NA

Multiple options identified

	Event	gear	Score (given)
a	Heatwave in the winter season	ice fishing	1
a	Heatwave in the winter season	rod	5

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1-5; from 1 – very similar, to 5 – very different species composition]

	Event	Answer	Score	Method
a	Heatwave in the winter season	: 5 sea trout on open water, instead of perch and pike under the ice	-997	G
b	Heatwave in the summer season	: 5 redirect the fishing down below the thermocline	-997	G
c	Severe icing (in the winter season)	No information	-998	NA
d	Algal bloom (in the summer season)	More about changing location (deeper), 5 choose other species (we fish pike & perch, sea trout)	-997	G
e	Extreme cloudburst (in the summer season)	No other species, more that we don't fish	-998	NA
f	Gale	: 2 sometimes; if it blows too hard and we are supposed to fish trout, then sometimes we have to go in and fish perch & pike instead	-997	G
g	Storm	Cancel	-998	NA

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

	Event	Answer	Score	Method
a	Heatwave in the winter season	3	3	O

b	Heatwave in the summer season	2, a b it worse	2	O
c	Severe icing (in the winter season)	3	3	O
d	Algal bloom (in the summer season)	2	2	O
e	Extreme cloudburst (in the summer season)	3	3	O
f	Gale	3	3	O
g	Storm	3	3	O

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1-5; from 1 – very unlikely, to 5 – very likely]

We release 98% of the fish; we are more concerned with the survival of the fish, e.g., during a heatwave survival becomes difficult; we don't want to fish pike when it's 30 degrees, then we'd rather fish sea trout that can survive better when we release it, since it goes down to colder depth

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	3	3	O
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	3	3	O
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

- Yes, I do have to change the setup of the fishing. With milder winters, I can't fish or might even take a regular job

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	4	NA
b	Heatwave in the summer season	NA	-999	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Answer	Score	Method
5, since I am chairman I have the mandate to represent the others	- 997	NA

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

	Event	Answer	Score	Method
a	Heatwave in the winter season	1 I do it for ice fishing; if no one falls in now, it is a miracle	1	O
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	2	2	O
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	1	1	O
f	Gale	1 (ordinary gale none, but strong gale)	1	O

g	Storm	2	2	O
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Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

	Event	Answer	Score	Method
a	Heatwave in the winter season	0	0	O
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	don't know, probably 2. (have broken parts on the boat twice because of it)	2	O
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	don't know, more if the boat sinks; is that a 3 then, don't know, depends on how big the damage is; in combination with storm then it becomes 3. Usually, you get away from a cloudburst. There is a risk; maybe 2	2	O
f	Gale	1	1	O
g	Storm	3	2	G

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
10 20k kr	10000	20000	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
20 600k kr	20000	6e+06	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

Change all the time according to the conditions; place, species and gear

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	NA	-999	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Answer	Score	Method
great similarity between boat guides (1)	1	NA

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

Easy for me to switch but ask what the customers want (e.g., Americans who are booked for ice fishing). Large differences from the normal are a problem, with bookings we do not know what we shall book (e.g., closed Nyköpingsån fishing due to heatwave). The customers also have to be flexible. 2 for those that are not dangerous; 1 for what is dangerous (cannot conduct activities)

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	1	NA
b	Heatwave in the summer season	NA	2	NA

c	Severe icing (in the winter season)	NA	1	NA
d	Algal bloom (in the summer season)	NA	2	NA
e	Extreme cloudburst (in the summer season)	NA	1	NA
f	Gale	NA	1	NA
g	Storm	NA	1	NA

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Answer	Score	Method
0 , if you take the extreme weather event to its peak, extreme cloudburst or storm, then the boat does not help; same if it is extreme heatwave, then we cannot go out 38 42 °C; in 2018 we could not book the fishing. Another parameter: our activity also depends on demand, and demand is made impossible during many of these extreme weather events	0	O

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score	Method
2	2	O

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score	Method
Sport fishing has great mobility; the accessibility to water. 2.	2	O

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
2 Yes	2	O

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Answer	Score Method	
Probably varies, yes it probably varies between fishers: some have a home harbor and are not so flexible; does that apply to management too do you think? don't know, generally not so much variation there	2	NA

Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

Type	Answer	Score Method	
a stress_injuries	5	5	O
b stress_gear	5	5	O
c stress_condition	4 (one does not want to fish if the fish are not doing well, it is also bad promotion for me as a fishing guide)	4	O

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Answer	Score Method	
it is so personal, some do not care at all, they just drink beer all day; we discuss things like this all the time, most are probably quite stressed by such situations	4	NA

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score Method	
I have to work, so I have no choice; everyone has the opportunity to work with something else who has two legs and two arms; I only guide, I have nothing I can lean back on. Is it 3? I would have to.	3	O

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score Method	
Would probably try to find solutions; sport fishing itself would certainly change, and then fishing guides would follow that change; there is probably some variation among guides depending on what one is specialized in	4	NA

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Answer	Score Method	
1. there are no possibilities; completely self financed. The bank would just laugh at one. The businesses are too small, and no one invests in this. Our whole industry is like that	1	O

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Answer	Score Method	
see answer above; the question was not asked, because the answers had already been made.	2	NA

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
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the world does not end without fishing guides; but we are ambassadors for how the development of fishing continues; it matters a lot how we handle fish & fishing in front of the general sport fishing mass, which is very large, 20% of Sweden's population; 5. important for a positive handling of fish and restoration of habitat. We are the only voice for the fish in the water. Important that there is a small scale tourist fishing industry, for a positive development for the fish	5	O
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Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
quite a lot; 3; influences so that more sport fishing equipment is sold in the nearby shops, hotel nights, everything around lunches, dinners, and so on	3	O

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
5. see answer to 27.	5	O

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
1. fish are not cows. We do not fish for food, almost none of us.	1	O

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27-30) vary between fishers in the group you represent

Answer	Score	Method
great variation 5.	5	O

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
5	5	O

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Answer	Score Method	
5	5	O

Q 34 - home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Answer	Score Method	
5, it creates positive memories; that is how one creates a relationship to places	5	O

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Answer	Score Method	
it is probably very unanimous among sport fishers; that is why one works with this; 1.	1	O

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Answer	Score Method	
100	100	O

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit

6000 8000 SEK/day and boat, generally the same for everyone	6000	8000	sek_day
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Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
oh, you don't want to know! what I do is so uneconomical; if the engine breaks then several 100,000 SEK go up in smoke, electric motor 50,000 SEK, sonar 50,000 SEK; is it rather 100,000 200,000 than 10,000 50,000 SEK? yes. Large part is expenses for material, applies also to the cars; 5 days a week fishing, 1 service day per week, 1 family day	50000	10000	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
maybe 1000 SEK/fishing day; this is the biggest expense besides food for a day	1000	1000	sek_day

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
1500 SEK for such a longer travel day	500	500	sek_day

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit

I always guide myself; there are some who sometimes take in an extra person; quite expensive, since it is salary and employer's contributions etc. (guess they get 1500 SEK, so in total it becomes 3000 SEK).

2000

3000

sek_day

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
a whole workday for service, so it costs what I earn on a fishing day; no, not for algal bloom, but in gale or storm there are many lines that must be fixed; twice I have broken the boat due to icing, the trim motor and the steering, 15,000 SEK per time.	NA	NA	sek_day

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
the biggest cost is the loss of income. Cancelled fishing days. A tourist usually does not rebook, especially if it is short notice change then I don't have time to get in something else. Guides for corporate events are at 30,000 SEK, or 60,000 SEK when we are out with 6 boats; if you miss that then the whole income disappears; loss of income also arises because people do not book when there is extreme weather.	NA	NA	sek_day

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Answer	Score Method
significant variation; 5.	5 NA

Q 45 - NA

Anything you want to add?

[free text description]

Answer	Score Method
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There may be disease among the fish during heatwaves, and that can have consequences for fishing for five years ahead; and algal bloom we do not know, but it does not feel good. We are completely dependent on the fish being well. How have perch and pike been doing this summer?

It was a bit weak when it was hot, but now they have started again. Probably quite a normal year. One adapts to the new. Sour that one cannot fish in the outer archipelago. It is an enormous difference from how it was when I was little, we would need supportive stocking of the pike stocks. First year with protection. I have not seen so much herring in 15 years! We have had incredibly little fish coming in to spawn earlier; when they last did test fishing at Mörkö, there was an 84 86% decline in spawning herring. Out last week, then there was herring in Furusundsleden, a lot of herring, proper schools. Summer herring. Don't know when I last saw that. Very positive! Will have great impact on the predatory fish stocks. Hope that the seals can move out again, so they don't have to hang around

Interviewee: I 2

2. name: teija aho. category: coastal fisherman id2 is a professional costal fisherman who operates in the north of bothnian bay (kalix). they fish in coastal waters with gillnets and traps (salmon traps and other traps for perch and bream), and they operate from boat at very short distances from their harbour. the target they mention is perch, bream, herring, salmon. they also run vendace fishery for roe production. the main fishing season is summer (from april to august). in winter they generally do not go out until spring comes. they can do sporadically some ice fishing. how they adapt: id2 fish has a structured succession of fishing styles within season. depending on the moment, they want to aim for something and may be forced to change. in the case they are fishing with nets (for pike, perch and herring?) they are flexible. their options to avoid hw and ab are to fish deeper, to avoid storms is to choose a different bay. hw especially affects perch, which die immediately. they notice more breams with hw. salmon traps are very rigidly managed and tied to a unique boat. salmon traps do not suffer from hw, but they do of storm, so if they must haul them up with a bigger boat, they get a penalty. vendace season is also rigid and short.

Q 2 - go_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1-5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Extremely warm winter we ice fish so little, therefore it is a bit difficult to answer. (2)	-999	G
b	Heatwave in the summer season	(3) Depends a little on what we are fishing	3	O
c	Severe icing (in the winter season)	1	1	O
d	Algal bloom (in the summer season)	Doesn't affect us much up here; locally there may be algal bloom in a bay. (4)	4	O
e	Extreme cloudburst (in the summer season)	3	3	O
f	Gale	1, not me at least with my small boat	1	O

g	Storm	1	1	O
h	Are there any of these in combination that affect how likely it is that you go out and fish?	Not really; it's the wind that is decisive for us	NA	NA

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

	Event	Answer	Score	Method
a	Heatwave in the winter season	3	-999	G
b	Heatwave in the summer season	2, Becomes worse, maybe not for all species, but they move to other areas, so you have to search elsewhere	2	O
c	Severe icing (in the winter season)	Hypothetical question, since we always have ice in winter; icing can happen in autumn, but we rarely fish then, so can't answer that	-998	NA
d	Algal bloom (in the summer season)	Hard to answer, doesn't occur very often	-998	NA
e	Extreme cloudburst (in the summer season)	3	3	O
f	Gale	It becomes harder to fish, but does not affect the actual catch; (2). Can affect a little since it's harder to fish	2	O
g	Storm	In a storm you can't fish, so catchability is zero	1	NA

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

	Event	Answer	Score	Method
a	Heatwave in the winter season	1	-999	G
b	Heatwave in the summer season	Somewhat different species if we fish in the same place e.g., a lot of bream; somewhat different (4).	4	O
c	Severe icing (in the winter season)	NA	-998	NA
d	Algal bloom (in the summer season)	NA	-998	NA

e	Extreme cloudburst (in the summer season)	1	1	O
f	Gale	1	1	O
g	Storm	NA	-998	NA

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

	Event	Answer	Score	Method
a	Heatwave in the winter season	3	-999	G
b	Heatwave in the summer season	1 Yes, then it gets quite a lot worse. Perch cannot tolerate lying in nets; they die almost immediately. Salmon in traps is a short fishing period and in deep water, so it doesn't get warm enough to affect them	NA	M
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	If there were to be strong algal bloom, which hasn't happened so far, then it would certainly affect negatively hypothetical.	2	NA
e	Extreme cloudburst (in the summer season)	NA	-998	NA
f	Gale	NA	-998	NA
g	Storm	NA	-998	NA

Multiple options identified

	Event	target	Score (given)
b	Heatwave in the summer season	perch	1
b	Heatwave in the summer season	salmon	3

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	Fish in deeper waters, maybe a middle value (3).	3	O
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	Fish in deeper waters, not difficult to avoid	4	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition 1–5; from 1 – not at all, to 5 – very much:

(3) Wind and heat will have long term impacts

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	3	NA
b	Heatwave in the summer season	NA	3	NA
c	Severe icing (in the winter season)	NA	-998	NA
d	Algal bloom (in the summer season)	NA	-998	NA
e	Extreme cloudburst (in the summer season)	NA	-998	NA
f	Gale	NA	3	NA
g	Storm	NA	3	NA

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Answer	Score	Method
1 Very little the conditions are about the same for everyone	1	O

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening]

	Event	Answer	Score	Method
a	Heatwave in the winter season	0	0	O
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	2 boat may sink	2	O
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	0 no major concern	0	O
g	Storm	2	2	O

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

	Event	Answer	Score	Method
a	Heatwave in the winter season	0) No boat used, the snowmobile can get stuck, but no damage, only extra work	-999	G
b	Heatwave in the summer season	More work cleaning gear from fouling, (1) if that can even be called damage.	1	O
c	Severe icing (in the winter season)	1	1	O
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	(1) If the boat sinks or the engine ends up under water (that's enough for one summer).	1	O
f	Gale	1	1	O

g	Storm	(2) Mostly to gear. It very rarely happens that we lose gear completely, but a small risk exists	2	O
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Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Tens of thousands of SEK, nets and such	10000	50000	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Several hundred thousand SEK	1e+05	5e+05	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

	Event	Answer	Score	Method
a	Heatwave in the winter season	(1) We might choose areas where travel on ice is easier.	1	O
b	Heatwave in the summer season	(1) Might change areas a little bit	1	O
c	Severe icing (in the winter season)	We fish so little during that season that it's hard to say	-999	NA
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	(0) We wait until the downpour is over and then go out; usually it only lasts a few hours.	0	O
f	Gale	(2) Then we may not go out at all.	2	O
g	Storm	(2) Then it's quite certain we won't go out.	2	O

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Answer	Score	Method
1	1	O

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Not so easy, rather difficult (2).	2	O
b	Heatwave in the summer season	(2) Quite difficult.	2	O
c	Severe icing (in the winter season)	can't answer	-999	NA
d	Algal bloom (in the summer season)	Quite easy (4).	4	O
e	Extreme cloudburst (in the summer season)	Doesn't affect much, so (not scored).	-999	NA
f	Gale	When it's windy, it's windy everywhere, so quite difficult (1) although you can always find some spot.	1	O
g	Storm	Impossible to fish at all (1).	1	O

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Answer	Score	Method
1	1	O

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score	Method
We only have one landing harbor, so it requires us to stay in the local area. (0)	0	O

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
Generally, we have quite a lot of mobility in terms of management; fixed gear locations are accessible fishing spots. Only for salmon fishing the license is tied to a specific boat, so you can't switch to a larger boat. For net fishing it doesn't matter. (Example given: switching to a larger and safer boat than the one tied to the salmon license in order to haul the salmon trap during gale, but then getting a penalty on the license).	999	NA

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
1 Depends on the fishing method. Salmon is the most restricted by management; it's the hardest to switch between species. Vendace (siklõja) too, because of the short season. At that time, we have no other alternative species to fish.	1	O

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Answer	Score Method	
1 same for everyone	1	O

Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

Type	Answer	Score Method	
a stress_injuries	Personal injury during fishing: 3	3	O
b stress_gear	Losing fishing gear: 2 not so stressful since it is quite rare	2	O
c stress_condition	Low quality of the fish in the catch: 2 possible to avoid so not very stressful	2	O

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Answer	Score Method	
4 It varies a lot; some don't dare to go out when it's just a little windy.	5	G

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score Method	
Personally: (5). For Bothnian Bay fishers in general: If a large part of the vendace season were stormy, it would be very hard to replace it with other income. Most have other jobs in winter, but fishing is such a significant part of income	1	G

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score Method	
5 We do it all the time, try to find new products and species to use. People must learn to eat more species	5	O

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Answer	Score Method	
1 If you want to borrow 1 million for a machine, you need to have 1 million in your pocket. Not many fishers have that	1	O

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Answer	Score Method	
5 Very large variation.	5	O

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
5	5	O

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
5 Vendace fishing is extremely important for Kalix, not only for jobs but also for the town's image	5	O

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
5	5	O

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
[5] Very important: vendace, salmon, and herring.	5	O

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27-30) vary between fishers in the group you represent

Answer	Score Method	
[1] No variation. It is very important for everyone, I dare say.	1	O

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
4 important	4	O

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Answer	Score Method	
[4] Important.	4	O

Q 34 - home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Answer	Score Method	
Hard to say for specific places. For example, Kalix archipelago what should I say? We live here, so yes, important.(4)	4	O

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Answer	Score Method	
[1] Not much variation. Most are very proud to be fishers; how much one fishes in practice may vary, but in terms of meaning, there are no major differences	2	G

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Hard to say. If we have 6 months without ice, we actively fish 1 third of the days. But we keep the gear in the water all ice free days.	50	100	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Impossible to calculate per day. Over 6 months? It's the four weeks of vendace fishing that make up the income. This varies. I haven't broken down exactly what the fishing itself gives, plus there's all the processing afterwards. Then you'd also have to deduct labor costs. Teija calculates roughly to get a value: 1.4 million SEK per year from vendace fishing. Plus some perch and bream. We haven't fished salmon for several seasons — the quota is fished up in 7 days [by those who have good fishing waters near the Torne River], so it's not worth setting the gear, since it takes a week to set and a week to haul.	1e+06	1500000	sek_season

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
I don't remember, it gets lumped together with all other repairs at Guldhaven. We buy 1 boat per season several hundred thousand SEK and a new trawl, 200k SEK, other stuff 100k SEK. Roughly 0.5 million SEK over two years.	1e+05	5e+05	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Don't dare estimate.	NA	NA	sek_day

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit

Marginal, since we have such short distances	NA	NA	sek_day
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Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
None for fishing itself. We bring in staff for processing [130 people). Finding staff is the limiting factor in fishing, especially when there is as much vendace as there is now	NA	NA	sek_day

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Cleaning traps takes some time, and it gets worse as the water gets warmer. Cleaning 10 traps took 0.5 h. Now we only have 20 traps in the water because of the perch, but when we fish the most, we have 100 traps. Then it means a lot of driving to clean them every three weeks; otherwise, they stay all season, or we clean once. Built ourselves from mink wire. Clean with a pressure washer. Same with bream traps	3	3	minutes_gear

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Things blow away. Doesn't cost much, but takes time to pick everything up when it's scattered across the bay.	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Answer	Score	Method
5	5	O

Q 45 - NA

Anything you want to add?

[free text description]

Answer	Score	Method
The striped perch are real stripes! They have found more in test fishing. Small ones, now frozen, which she will send to us. 234 perch in the test fishing, 9 33 cm. We have a lot of perch here now, never been so many on rod fishing — in a 1 hour round we get 10 perch of eating size. Why do you think? Now that the vendace stock is so good, they have a lot of food, and they grow well when it's warm. Seems to be good recruitment too, since we caught so many small ones. We had the nets here in Långviken, along a gradient from open sea to inner bay: mostly bream in the innermost net, and mostly perch in the outermost net. The largest perch were farthest out.	- 999	NA

Interviewee: I 3

3. name: dennis and lars-göran. category: coastal fisherman id3 is professional coastal fishermen in southwest bothnian sea (norrsundet). they fish with gillnets and traps. traps are only used for salmon. the main target is herring, whitefish, salmon, perch, and eel. they also run a restaurant and have several fish trucks. the herring fishing takes place a few hundred meters from the harbor. they mention whitefish, salmon, perch in the local area, and in the gulf of gävle. their average distance travel is about 1 hour (assuming 10 knots it is about 20 km) and the longest distance travel is 3 hours (assuming 10 knots it is max 55 km) the main fishing season is summer (from april to august). in winter they generally do not go out until spring comes. how they adapt: id3 can adapt when fishing with nets to hw by avoiding hottest hours, performing shorter sets and using cooling boxes. can also adapt to hw, ab and gales by changing fishing location. to avoid hw and ab they can go further out, the opposite for gales. in any case they can move on short distances and can cope with gales only in shallow waters, they don't go deep and with storms they don't go out at all. their spatial mobility is mainly affected by the boat (12m boat) and infrastructure (they want to go back home) on second instance by management that makes possible to fish in proximity (over a coastline spanning of about 60 km in aerial distance). there is not the same flexibility for salmon traps, which are fixed.

Q 2 - go_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Depends on ice. Boats are usually hauled up in winter. If forecast shows winter is over, we might launch. Unlikely to put in and haul up again. Might use the small boat if open water, but ice may return. Large boats remain on land. =1 regardless.	-999	NA
b	Heatwave in the summer season	Then we fish, just go further out where it's cooler. =5.	5	O
c	Severe icing (in the winter season)	5 (icing can happen in spring).	-999	G

d	Algal bloom (in the summer season)	Rare up here, but still fish. =5.	5	O
e	Extreme cloudburst (in the summer season)	If gear is already out, we go anyway. If not, we wait. =3.	3	O
f	Gale	No, we stay home. =1.	1	O
g	Storm	No, we stay home. =1.	1	O
h	Are there any of these in combination that affect how likely it is that you go out and fish?	Wind has the greatest impact here. Wind direction matters too (this August mostly W & NW winds, cooling the water – no point fishing perch then, they just stop moving).	NA	NA

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Not relevant	-999	NA
b	Heatwave in the summer season	Heatwave makes it worse. =2.	2	O
c	Severe icing (in the winter season)	not relevant	-999	NA
d	Algal bloom (in the summer season)	For perch, it gets worse (=2). Same for herring, due to warmth.	3	M
e	Extreme cloudburst (in the summer season)	Doesn't affect catchability (only whether we go out). =3.	3	O
f	Gale	Yes – after a gale from the right direction, herring can blow in, salmon may swim more too (E, NE in summer). =4. From W & NW it gets worse instead.	NA	M
g	Storm	Same, just stronger wind.	NA	NA

Multiple options identified

	Event	target	Score (given)
d	Algal bloom (in the summer season)	perch	2
d	Algal bloom (in the summer season)	herring	2
f	Gale	herring	999

f	Gale	salmon	999
g	Storm	herring	999
g	Storm	salmon	999

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Not relevant.	-999	NA
b	Heatwave in the summer season	very similar (1)	1	O
c	Severe icing (in the winter season)	not relevant	-999	NA
d	Algal bloom (in the summer season)	not directly (1)	1	O
e	Extreme cloudburst (in the summer season)	no (1)	1	O
f	Gale	1	1	O
g	Storm	1	1	O

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

	Event	Answer	Score	Method
a	Heatwave in the winter season	not relevant	-999	NA
b	Heatwave in the summer season	much worse, must get them quickly into cooling boxes (1)	1	O
c	Severe icing (in the winter season)	not relevant	-999	NA
d	Algal bloom (in the summer season)	no, just from the heat	3	NA
e	Extreme cloudburst (in the summer season)	no (3)	3	O

f	Gale	no (3)	3	O
g	Storm	no (3)	3	O

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1–5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	Only b. Go out before midday heat. Early mornings for herring, perch later in day. Adjust by hauling nets earlier. =5.	5	O
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

1

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	NA	1	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	1	NA
e	Extreme cloudburst (in the summer season)	NA	1	NA
f	Gale	NA	1	NA

g	Storm	NA	1	NA
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Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Answer	Score	Method
1 in Bothnian sea	1	O

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	rarely go out but sudden events possible	1	NA
g	Storm	don't go out	-998	NA

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	NA	0	NA

c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	0, though nets can get dirty.	0	O
e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	Could damage or destroy gear/boat, or cause minor damage	999	NA
g	Storm	Could damage or destroy gear/boat, or cause minor damage	999	NA

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
50,000 SEK/year for nets that break, or propellers when blown onto land/rocks	1	50000	sek_year

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Rare for boats – then we wouldn't be out. But a salmon trap can be destroyed in a summer storm, costing 200,000 SEK	50000	2e+05	sek

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0-2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	Yes, set in other places, change a lot. =2. Salmon gear is fixed at paid spots all summer, so they stay.	2	O
c	Severe icing (in the winter season)	NA	-999	NA

d	Algal bloom (in the summer season)	In bays, same as in heatwaves	2	NA
e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	2. Move to less windy spots. For herring it's harder, so maybe stay home.	2	O
g	Storm	Don't go out at all, follow forecasts. Our boat is 12 m, so never in storms	-998	NA

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Answer	Score	Method
Uniform. 1.	1	O

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	Works – go further out (usually warmest inshore). =5.	5	O
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	If heat lasts long, algal bloom spreads and worsens fishing. =2.	2	O
e	Extreme cloudburst (in the summer season)	Yes, possible.	4	NA
f	Gale	Yes, but usually means staying home	4	NA
g	Storm	Storms are hard to avoid.	2	NA

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Answer	Score	Method
Yes, can move around for herring, whitefish, perch. Only salmon gear is fixed. Our operations use active gear, so we can relocate, but only short distances – catch must be landed quickly. (Older generations even traveled to Trosa/Grisslehamn.)	999	NA

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score	Method
No – no infrastructure elsewhere, must return home. =0.	0	O

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score	Method
Not well informed. Probably possible in Gävle Bay (we have exemptions locally). =1.	1	O

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score	Method
0. Few commercial species here, license not species-specific	0	O

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Answer	Score	Method
Same situation for all. =1	1	O

Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

	Type	Answer	Score	Method
a	stress_injuries	1, not stressed about that.	1	O
b	stress_gear	Yes – after summer storms worried salmon traps may blow away. =5.	5	O
c	stress_condition	Can happen if winds last days, then quality is lost, fish becomes worthless. Usually not stormy that long. =3.	3	O

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Answer	Score	Method
Fishers are calm and experienced, know when to go out. =1.	1	O

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score	Method
Yes – we already do many other things. Impossible to live solely off fishing on the Norrland coast. Must diversify, requires entrepreneurship. =5	5	O

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score	Method
Always striving to improve, never want to stop developing. Every time we smoke fish, want it better. Always want renewal. =5	5	O

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

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Answer	Score Method	
1. Not a thriving industry. We've thought about new boats for better work environment, but economically impossible with so little fish	1	O

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Answer	Score Method	
Not all alike – some more ambitious than others, depends on vision. Some are “fair-weather fishers,” others try to make it work year-round. =5.	5	O

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
Very important. We want it to live on. Almost dying out, barely any succession in families. In Gävle municipality soon only us left. =5. That's why I started – to continue and develop what earlier generations built. History carries weight.	5	O

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
Very important, =5. We're the largest employer here. In July we had 50 on payroll. Coastal fishing benefits many. Attracts 40–50,000 tourists annually. They fuel up, shop at local stores. Huge significance for the whole community.	5	O

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
5. Doesn't have to be me, but important that I'm the youngest investing – inspires cousins to invest too. Many long-time fishers easily deliver fish to Bergmans.	5	O

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
5. Very important for Swedish food production.	5	O

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Answer	Score Method	
Very similar. That's why they continue, try to recruit new ones. But hard since chances are poor, little money. =1.	1	O

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
5	5	O

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Answer	Score Method	
5. Now it's my whole identity (used to play hockey, now entrepreneurship in fishing defines me).	5	O

Q 34 - home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Answer	Score Method	
Very important – many specific places with memories. As important as self-image. =5	5	O

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Answer	Score	Method
Quite similar – everyone loves being outdoors, values the freedom. Calm water, sun shining – few jobs beat that!	2	NA

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Varies, 120–150 days of net fishing.	120	150	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Hard to say. Some days 10–15,000 SEK, other days 0 if seals eat everything	10000	15000	sek_day

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
About 200,000 SEK. 100–150,000 in damages. A lot of time spent on repairs, much voluntary work, patching holes in winter	150000	2e+05	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

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Answer	Score - low (given)	Score - up (given)	Unit
Several thousand SEK. Longest trips 2–3 hours out: 3,000–5,000 SEK/day. Short trips about half. Sometimes just outside the restaurant here	1500	2500	sek_day

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
See previous	3000	5000	sek_day

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Rare – usually only if a lot of fish. Then about 200 SEK/hour, unless pensioners help voluntarily.	0	200	sek_hour

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
20–30 hours. Salmon traps: ~20 h each to clean at season end, sometimes more. Warm July brings more growth. In 2018 it was so warm that when we hauled a trap, with so much fouling, we nearly sank when a waterspout hit	20	30	hours_event

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit

Mostly covered – repairs, gear blown ashore.	NA	NA	not_reported
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Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Answer	Score	Method
No, not much variation. We use the same type of gear	2	NA

Q 45 - NA

Anything you want to add?

[free text description]

Answer	Score	Method
No, I think everything's covered. Biggest problems are seals and industrial fishing. With so few fish, seals get aggressive	- 999	NA

Interviewee: I 4

4. name: kjell mahlöf. category: archipelago fisherman id4 is a resident of the archipelago in the northern proper baltic (hasselö, tjust archipelago). archipelago fisherman cannot sell their catchers. in the past, there was a very diversified fishing community with various types of nets for coastal and freshwater species. nowadays they continue to fish with nets for perch, herring, whitefish. he mentions also the importances of being able to catch fish with rods, but in the second interview he only refers to fishing with nets. fishing mostly happens during summer (april to august), winter fishing is less common but can happen. he mentions that if the water is free from ice they can cast nets also in winter. how they adapt: id4 is not economically dependent on fishing and in general tend to do it only when there are good conditions. he can adapt to hw by performing shorter (not overnight) sets. can also adapt to hw, ab and storms by changing fishing location. to avoid hw and ab they can go further out, the opposite for storms. in any case they can move on very short distances. their spatial mobility is mainly affected by its assigned fishing lot.

Q 2 - go_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Quite likely, if the water is open we'll go out, 4	4	O
b	Heatwave in the summer season	2. Depends on how it affects winds and algal blooms; this year we had huge problems with algal blooms (1 month).	2	O
c	Severe icing (in the winter season)	No problem for us, too far south.	-999	NA

d	Algal bloom (in the summer season)	1	1	O
e	Extreme cloudburst (in the summer season)	We can get to the boathouse, but from there out onto the lake it's more difficult; 1, if we don't have gear in the water. But we check the weather forecast very carefully before putting in the gear.	1	O
f	Gale	1	1	O
g	Storm	1	1	O
h	Are there any of these in combination that affect how likely it is that you go out and fish?	Strong winds and high water levels.	NA	NA

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	2	NA
b	Heatwave in the summer season	Yes, in the past when perch existed, they liked warmth, but now there are no perch, so the effect is not so big on the species we fish for	3	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	It's devastating, depressing to live around this algal soup, grandchildren can't swim, it's an imbalance in the Baltic Sea ecosystem, a sad new phenomenon. We try to deal with it when it drifts ashore. It becomes a white coating on the bottom when it sinks. We're worried about algal toxins (cows can break them down somewhat, but single-stomached animals can't; affects the liver). 1.	1	O
e	Extreme cloudburst (in the summer season)	No, 3	3	O
f	Gale	A little worried about strong winds since we don't have big boats.	2	NA
g	Storm	Same with storms.	2	NA

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Can't answer.	-998	NA
b	Heatwave in the summer season	Can't answer.	-998	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	We don't really fish much then; would guess the fish flee from the algal bloom, 5	5	O
e	Extreme cloudburst (in the summer season)	Can't answer	-998	NA
f	Gale	Hard to say, we don't fish much then	-998	NA
g	Storm	Hard to say, we don't fish much then	-998	NA

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Don't think it affects condition, 3.	3	O
b	Heatwave in the summer season	hen the water is warmer, fish flesh becomes looser (especially fattier fish), 2.	2	O
c	Severe icing (in the winter season)	we don't fish then	-999	NA
d	Algal bloom (in the summer season)	we don't fish then	-998	NA
e	Extreme cloudburst (in the summer season)	we don't fish then	-998	NA
f	Gale	we don't fish then	-998	NA
g	Storm	we don't fish then	-998	NA

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1-5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	If perch turn yellow and bad from the heat, we can do a day-set (not overnight). Have also set nets around a shoal and driven fish inside, which improved quality. We don't have ice machines, so we can influence it a little	3	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

Algal bloom doesn't need more comment, it's a big problem we've already discussed. Other factor is wind direction, NW is bad, then the fish disappear. Big impact. This spring it locked in at N & NW with near gale winds for several weeks

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	NA	-999	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	5	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	5	NA
g	Storm	NA	5	NA

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2-7) vary between fishers in the group you represent? 1-5; from 1 – not at all, to 5 – very much

Answer	Score	Method
Among subsistence fishers there's quite strong agreement; we communicate fairly well with each other, 2	2	O

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0-2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

With strong winds, you adapt as an islander. I also do forestry, and if weather is bad with fog and strong winds, you plan activities so as not to risk injury. You adapt to weather

	Event	Answer	Score	Method
a	Heatwave in the winter season	no worry	0	NA
b	Heatwave in the summer season	no worry	0	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	avoid eating fish caught during bloom	0	NA
e	Extreme cloudburst (in the summer season)	no worry, just wear the right clothes	0	NA
f	Gale	slipping, breaking a leg from the rocking	1	NA
g	Storm	slipping, breaking a leg from the rocking	1	NA

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0-3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

	Event	Answer	Score	Method
a	Heatwave in the winter season	0	0	O
b	Heatwave in the summer season	with low water (running aground), 1	1	O
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	0	0	O

e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	Gear can drag and be damaged, very rare; boat damage when docking can be critical, minor repairs, 1.	1	O
g	Storm	Gear can drag and be damaged, very rare; boat damage when docking can be critical, minor repairs, 1.	1	O

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Propeller replacement is quite heavy, costs 2,500 SEK. Hull repair we fix ourselves, a couple hundred SEK	200	2500	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Engines and boats are insured; we must cover gear ourselves. A few thousand SEK more than the 2,500.	2500	5000	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0-2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

	Event	Answer	Score	Method
a	Heatwave in the winter season	0	0	O
b	Heatwave in the summer season	0 Only with algal bloom. If we want whitefish in summer during heatwave, they move deeper (15 m), =1. Move a few hundred meters	1	O
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	2 then we do not go fish	2	O

e	Extreme cloudburst (in the summer season)	Don't go out to set nets, but we do collect them if already out	1	NA
f	Gale	We sometimes set in more sheltered bays less exposed, =2.	2	O
g	Storm	We sometimes set in more sheltered bays less exposed, =2.	2	O

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Answer	Score	Method
1, no	1	O

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

	Event	Answer	Score	Method
a	Heatwave in the winter season	no need	-999	NA
b	Heatwave in the summer season	Drive grandchildren to another bay; no, we don't change fishing spot, we stop fishing	1	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	Stop fishing	1	NA
e	Extreme cloudburst (in the summer season)	1, no, we stop completely	1	O
f	Gale	4, switch bays	4	O
g	Storm	Don't go fishing	1	NA

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Answer	Score	Method
1, somewhat	1	O

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
Yes, but we don't, too much hassle. Since we're not economically dependent, we can skip fishing. =2.	2	O

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
We have common fishing waters (a cooperative). Fishing lots have become permanent, so I stick to my lot. =0 (within cooperative).	0	O

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
Eel fishing ban; voluntary fishing stop Mar 15–May 15 in some spawning bays. We must restore predatory fish stocks. We think fish shelters (we've built 10 over 10 years) have limited effect at such low stock levels.	999	NA

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Answer	Score Method	
Requires a vehicle/tractor, not all have it, but roughly consistent, =2.	2	O

Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

	Type	Answer	Score	Method
a	stress_injuries	not much, but in winter it's cold, windy, if motors fail then big problem, 3.	3	O
b	stress_gear	1	1	O
c	stress_condition	1	1	O

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Answer	Score	Method
Probably fairly consistent, =2.	2	O

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1-5; from 1 – very unlikely, to 5 – very likely]

Answer	Score	Method
Not Asked	- 999	NA

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1-5; from 1 – very unlikely, to 5 – very likely]

Answer	Score	Method
Thinking about whether nets are best; with traps the fish are less affected. Just thoughts so far. =2.	2	O

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1-5; from 1 – very difficult, to 5 – very easy]

Answer	Score	Method
Not Asked	- 999	NA

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Answer	Score	Method
Most are conservative, professionals are more innovative. In my group, low innovation.	2	NA

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
5. We’ve lost local competence when people die and take knowledge with them (e.g., neighbor who knew when/where to catch turbot, burbot).	5	O

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
I can’t sell/deliver fish, only the professional fisher can. More cultural—nice to have fish on the table. =1.	1	O

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
Hard to judge. There have been big changes recently – many have passed away or disappeared, and some younger people have moved in who have the opportunity to fish but haven’t started yet, they’re still renovating. At our fishing meetings people do show up – they’re not uninterested. People are interested in fishing, but to what extent they actually fish or communicate about it varies. It creates a unifying factor among us subsistence fishers who belong to the village association. Those who own a joint property with fishing rights and waters – and then there are the summer guests, but I have no idea about them.	3. 3	O

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
Generational gap. We like fried herring, but grandchildren want hamburgers. Son hates flounder (too much growing up). Daughter better. Have to teach them every time how to debone herring. 5. Also for preparedness—resilience groups forming on islands (root cellars, handling power cuts).	5	G

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Answer	Score	Method
Most people think roughly as we have just said.	2	NA

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
5. During test fishing we had wonderful mornings lifting nets—that's what you live on.	5	O

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Answer	Score	Method
Not so much the fishing, but managing it (chair of fishing association). =3.	3	O

Q 34 - home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Answer	Score	Method
There are several places we often go. Has some meaning, about 3	3	O

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Answer	Score	Method
Can't say, emotional things	-	NA
	998	

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Not much lately, maybe 10 days.	10	NA	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
5k SEK	NA	5000	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit

Significant; 20,000 SEK yearly, much for boats, transport to land, island transport. About 5,000 SEK/year.	5000	20000	sek_season
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Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
When cleaning fish, takes a lot of time.	NA	NA	not_reported

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
I think we've gone through most of it. Water levels: your report mentions it, and many are concerned about low water levels, which indeed happened; but high water is just as problematic – wading out to the boathouse in waders.	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Answer	Score	Method
There are some who have fishing rights but don't fish at all, and then some who fish like I do, just now and then. Then there are those who have leased out their fishing lot, and in those cases there is a subsistence fisher who fishes quite intensively	5	NA

Q 45 - NA

Anything you want to add?

[free text description]

Answer	Score	Method
Regarding innovation: we're interested in all cooperation with experts who can help us restore ecological balance in the Baltic Sea. There is interest moving forward. What I pick up is that we must do something about the algal blooms. Maybe find a way to collect them with boats; there is centrifuge competence in Sweden	- 999	NA

Interviewee: I 5

5. name: lars ljunggren. category: recreational fisherman id5 represents the organization sportfiskarna, which is active in all sweden. he is based in stockholm. the members of sportfiskarna are anglers that mostly fish for pike, perch and zander. additionally, fishing for sea trout and trolling for salmon is common. they fish in lakes, streams and coastal areas. they can use boats in coastal areas and lakes, or just fish from the shore. in winter they also do ice fishing on lakes and generally they do not use boats until the ice is completely gone. the members practice both catch-and-release and keeping the catch. recreational fisherman can move around different fishing spots. how they adapt: sportfisherman are diverse but have in common to fish exclusively for fun. they tend to fish only when it is a pleasant experience. they can cope to some extent with extreme events by adapting their location if they want to fish on the coast (going deeper or finding sheltered places) or move to lakes. their spatial mobility has virtually no limits.

Q 2 - go_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	If it's otherwise a good winter, then it's just nice weather. But if it's a mild winter, then you can't go out. Is ice fishing the dominant form? Few go out in boats? On the big lakes. and at Simrishamn there is some trolling from boats. If we think about ice fishing? Absolutely, =1 if the ice isn't there or is dangerous to walk on.	1	O

b	Heatwave in the summer season	Yes, but in another way. When it's hot, fishing in running waters (salmon, trout, grayling) is closed. Perch and zander fishing only gets better in warm weather. Few target pike in a heatwave. But if you booked a salmon-fishing week, it simply gets canceled. Zander & perch = 5; salmon & trout = 1; pike = 2 (or 1, discussed with another angler present). But "vacation fishers" will probably go out regardless. Pike fishing is usually in spring & autumn; high temps affect survival.	NA	M
c	Severe icing (in the winter season)	Big impact.	1	NA
d	Algal bloom (in the summer season)	Less attractive, but still possible to fish; =2.	2	O
e	Extreme cloudburst (in the summer season)	People hold back; it's a comfort factor. Most stay indoors	2	NA
f	Gale	Yes, people hold back; most stay home, except the really keen ones who may go out in sheltered spots	2	NA
g	Storm	No, impossible to fish.	1	NA
h	Are there any of these in combination that affect how likely it is that you go out and fish?	Combination of gale/storm and heavy rain – Even more unpleasant, a negative synergy effect	NA	NA

Multiple options identified

	Event	target	Score (given)
b	Heatwave in the summer season	zander	5
b	Heatwave in the summer season	perch	5
b	Heatwave in the summer season	trout	1
b	Heatwave in the summer season	salmon	1
b	Heatwave in the summer season	pike	2

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

	Event	Answer	Score	Method
a	Heatwave in the winter season	If there's no ice, you can't catch them with an ice rod, and you don't put in a boat just for a short ice-free spell.	-997	NA
b	Heatwave in the summer season	species that dislike heat (salmonids) and even pike become harder to catch; perch and zander improve	NA	M
c	Severe icing (in the winter season)	Heavy icing is rare, and few troll in sub-zero temps – marginal effect on fish	2	NA
d	Algal bloom (in the summer season)	Don't know	-998	NA
e	Extreme cloudburst (in the summer season)	No.	3	NA
f	Gale	Makes fishing more difficult, but doesn't affect catchability	2	NA
g	Storm	Nobody goes out in storms, so we don't know if fish bite then	-998	NA

Multiple options identified

	Event	target	Score (given)
b	Heatwave in the summer season	zander	4
b	Heatwave in the summer season	perch	4
b	Heatwave in the summer season	trout	2
b	Heatwave in the summer season	salmon	2
b	Heatwave in the summer season	pike	2

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

	Event	Answer	Score	Method
a	Heatwave in the winter season	No, very similar	1	NA
b	Heatwave in the summer season	No, you fish where you fish, so you get the species present there. But in mountain areas, if you target trout or grayling but the system also has pike and perch, you might get more of those instead. From a boat it's usually small differences	2	NA

c	Severe icing (in the winter season)	No, icing is rare	-998	NA
d	Algal bloom (in the summer season)	No	1	NA
e	Extreme cloudburst (in the summer season)	No	1	NA
f	Gale	Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season	999	NA
g	Storm	Not directly, but on the coast with prolonged winds you can get upwelling; perch may move closer inshore, and trout & whitefish may appear shallower than normal. Seen within a season	999	NA

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

	Event	Answer	Score	Method
a	Heatwave in the winter season	no	3	NA
b	Heatwave in the summer season	Yes, species that thrive in cooler waters get poorer condition	2	NA
c	Severe icing (in the winter season)	NA	-998	NA
d	Algal bloom (in the summer season)	No (guess)	3	NA
e	Extreme cloudburst (in the summer season)	no	3	NA
f	Gale	no	3	NA
g	Storm	no	3	NA

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1-5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA

b	Heatwave in the summer season	No, nothing can be done. If fish are in poor condition, there's nothing you can do. Doesn't really apply to sport fishing since fish aren't trapped in gear	1	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

Greatest effect is probably on Arctic char (though not a major sportfishing species). Other species may move in: grayling, trout, whitefish, perch & pike. Biggest sportfishing impact from extreme weather is likely on char. On the coast, fish can always go deeper. Does it affect how you fish? Yes – e.g., trout anglers (spring & autumn from shore) may need to fish deeper or at different times. On Gotland, sea trout fishing is big – if warming lasts longer, the season gets shorter until it cools again.

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	NA	3	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Answer	Score Method
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Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening]

	Event	Answer	Score	Method
a	Heatwave in the winter season	for those fishing then, big concern – must assess ice conditions if a winter heatwave occurs	2	NA
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	If someone is foolish enough to go out, some risk, but few do	1	NA
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	0	0	O
g	Storm	Surprise storms can be worrisome 2	2	O

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

	Event	Answer	Score	Method
a	Heatwave in the winter season	0	0	O
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	1 (but most boats are stored then)	1	O
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	could sink a boat if unlucky, 3	3	O
f	Gale	no	0	NA
g	Storm	yes could sink a boat (3)	3	O

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Anything from 10,000 SEK to 1 million SEK – these are expensive toys; hard to say	10000	50000	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Up to half a million SEK, not uncommon. For new trolling setups, even more. Sportfishing boats cost 50–300,000 SEK, sometimes double	50000	1000000	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0–2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

	Event	Answer	Score	Method
a	Heatwave in the winter season	2 – if the ice is unsafe, you don't go out.	2	O
b	Heatwave in the summer season	Mixed	NA	NA
c	Severe icing (in the winter season)	NA	-998	NA
d	Algal bloom (in the summer season)	Maybe 1 – switch to another bay; if bloom is everywhere, then skip.	1	O
e	Extreme cloudburst (in the summer season)	2 – you stay indoors	2	O
f	Gale	If sheltered, you can fish a little, =1, unless fully blown in.	1	O
g	Storm	Same	1	O

Multiple options identified

	Event	target	Score (given)
b	Heatwave in the summer season	perch	0
b	Heatwave in the summer season	salmon/trout	2

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Answer	Score	Method
Depends on type of fishing – varies a lot, =5.	5	O

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Hard, scale is large.	2	NA
b	Heatwave in the summer season	Hard	2	NA
c	Severe icing (in the winter season)	NA	-998	NA
d	Algal bloom (in the summer season)	Easier – could switch to a lake if planning coastal fishing	4	NA
e	Extreme cloudburst (in the summer season)	Hard	2	NA
f	Gale	to some extent – find a sheltered spot	3	NA
g	Storm	no	1	NA

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Answer	Score	Method
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Most can be somewhat mobile. Some are keen enough to travel far. 1 O
=1.

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
For mobile sport fishers, infrastructure is crucial. It exists, but not always accessible for everyone	1	NA

Q 18 - management__space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
2. But you may need to buy a fishing license for new spots. On the coast it's free, which allows a lot of freedom.	2	O

Q 19 - management__species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
Some restrictions. On the coast, fairly free. But salmon, trout & grayling are closed during spawning, or when it gets too warm (quite common). =1.	1	O

Q 20 - unc__15__19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Answer	Score Method	
Quite a lot – some don't fish from boats at all, some do.	4	NA

Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

	Type	Answer	Score	Method
a	stress_injuries	Not really – only extreme cases (climbing steep riverbanks, pushing	1	NA
b	stress_gear	no	1	NA
c	stress_condition	no	1	NA

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Answer	Score	Method
some variation, but less than other questions	2	NA

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score	Method
not asked	- 999	NA

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score	Method
I'm probably quite creative, I think so. =4.	4	O

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Answer	Score	Method
not asked	- 999	NA

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Answer	Score	Method
Varies greatly (only for Q24).	4	NA

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
Relatively important – nice if the next generation also enjoys fishing. Fairly important. =4.	4	O

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
Generally, not a strong link. But in rural areas, sport fishing can keep the village shop open due to visiting guests in summer. In rural areas it has real significance	5	NA

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
Some drive among those eager to show off successful fishing. But being part of that community doesn't really require fishing.	999	NA

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
In between. =3.	3	O

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Answer	Score Method	
Probably quite a lot.	4	NA

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
Quite significant – give it a 4.	4	O

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Answer	Score Method	
In between. =3.	3	O

Q 34 - home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Answer	Score Method	
Contributes strongly, at least a 4.	4	O

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Answer	Score Method	
Doesn't vary much. The place/environment matters, of course it varies somewhat, but everyone values getting out to beautiful places. =2.	2	O

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Not in SCB's survey. Average angler maybe 5–10 days, but some 50–100 days. More than 100? Yes, HC!	5	100	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Servicing a boat: at least 5,000 SEK and up. Without a boat: maybe service a reel every couple of years, ~100 SEK	100	5000	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Varies hugely. Trolling anglers burn a lot. Ordinary perch & pike anglers maybe 30–50 L/trip on average	30	50	liter_trip

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Maybe 50 L for a trolling angler. Boats are efficient, so 20–50 L/trip	20	50	liter_trip

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
NA	NA	NA	not_reported

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Some make their own lures/jigs/wobblers – for them, that’s half the fishing time, usually done in winter. Affected by algal bloom? No	999	999	hour

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Yes – e.g., if you booked a salmon fishing week and they close fishing, you’re stuck with costs for a cabin if it can’t be rebooked. Usually you can rebook, but I don’t know the cost.	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Answer	Score	Method
5	5	O

Q 45 - NA

Anything you want to add?

[free text description]

Answer	Score	Method
I think you've captured most of it. The most likely major impacts are wind – if extreme weather is prolonged, it affects access. And if you fish salmonids, a summer heatwave affects both fish and fishing opportunities. Same with winter ice fishing.	- 999	NA

Interviewee: I 6

6. name: lars-uno. category: pelagic trawlers id6 fish pelagic fish both in the baltic sea and the north sea. in the baltic sea, the fishing targets herring/ sprat and sharp-sprat, and the catch is sent to fishmeal. in the baltic sea, lars-uno fishes from the southern baltic up to the bothnian sea (at the level of sundsvall). they can occasionally fish in outermost regions (shetlands). the main fishing season is from august to april (but the season can extend longer into spring and summer), and the current law does not establish any limitation regarding time at sea (e.g.: seasonal ban, weekend ban). their activity is limited by the quota. in winter their activity in the baltic might be affected by ice coverage. how they adapt: pelagic trawlers have a great spatial mobility because of their large size vessels. their primary method to adapt to weather is to move around: they may switch from an area to another when they know the climatic conditions (i.e.: water temperature) are good. in the case of local weather is not good they stay in port. they have efficient cooling systems to deal with hot weather.

Q 2 - go_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	5	5	O
b	Heatwave in the summer season	1; wait a minute, spring when the weather changes; in July when it's hottest, the fish are scattered, so there's no point in going out; say 2.	1.5	G
c	Severe icing (in the winter season)	we avoid going out, 1.	1	O
d	Algal bloom (in the summer season)	5	5	O
e	Extreme cloudburst (in the summer season)	5	5	O
f	Gale	4 in the Baltic Sea at least; if we are going to consume the fish in the North Sea, then we wait because the fish can spoil , but in the Baltic Sea we don't fish for human consumption	NA	M

g	Storm	1, we have the quota so we don't need to risk the crew.	1	O
h	Are there any of these in combination that affect how likely it is that you go out and fish?	Icing & high winds, then it's 1	NA	NA

Multiple options identified

	Event	area	Score (given)
f	Gale	Baltic	4
f	Gale	NorthSea	1

Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

	Event	Answer	Score	Method
a	Heatwave in the winter season	It can go both better and worse, but it has an impact; for the same weather phenomenon, it can be both better and worse; weather changes affect catchability, as do moon phases and the time of year.	3	NA
b	Heatwave in the summer season	in the North Sea, herring fishing is good in the summer; in August in Skagerrakk, we saw that when the water became warm, we got large quantities of herring at Skagen when the temperature was 20-21°C; But in the Baltic Sea, the fish spread out when it gets warm. 2	NA	M
c	Severe icing (in the winter season)	We can't load the boat when it's icy. 1.	1	O
d	Algal bloom (in the summer season)	1	1	O
e	Extreme cloudburst (in the summer season)	Doesn't affect 3.	3	O
f	Gale	3. It can sometimes, but no 3	3	O
g	Storm	it can be harder for us to handle the boat and get where we want to go, so it does have an impact. 2.	2	O

Multiple options identified

	Event	area	Score (given)
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b	Heatwave in the summer season	Baltic	2
b	Heatwave in the summer season	NorthSea	4

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

	Event	Answer	Score	Method
a	Heatwave in the winter season	1	1	O
b	Heatwave in the summer season	NA	1	NA
c	Severe icing (in the winter season)	1	1	O
d	Algal bloom (in the summer season)	1	1	O
e	Extreme cloudburst (in the summer season)	1	1	O
f	Gale	1	1	O
g	Storm	1	1	O

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

	Event	Answer	Score	Method
a	Heatwave in the winter season	2, because the boat moves and the fish breaks	2	O
b	Heatwave in the summer season	1 same thing, in Lake Norske it becomes a problem because the boat moves and the fish breaks when it is warm	1	O
c	Severe icing (in the winter season)	3	3	O
d	Algal bloom (in the summer season)	3	3	O
e	Extreme cloudburst (in the summer season)	3	3	O

f	Gale	1	1	O
g	Storm	1	1	O

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1-5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	We cool with RSV systems, so it stays in good quality. 5	5	O
b	Heatwave in the summer season	We cool with RSV systems, so it stays in good quality. 5	5	O
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	You can go ashore and unload what you have. 3	3	O
g	Storm	You can go ashore and unload what you have. 3	3	O

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

4, we saw that mackerel migrate, but the funny thing is that they are starting to come back again, as they did 20 years ago. I don't know if this has anything to do with the heat, or if it's a cycle; mackerel have been north of Shetland, but now it seems that they are back in Norway again; perhaps they will return to how it was 20-25 years ago; In the Baltic Sea, I think cod has been affected because it has gotten warmer. Thirty-five to forty years ago, the entire Baltic Sea was frozen, but sprat and herring have increased in the last five years. Smelt has also recovered because it has gotten warmer. Yes, it has affected us.

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	4	NA
b	Heatwave in the summer season	NA	4	NA
c	Severe icing (in the winter season)	NA	-999	NA

d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Answer	Score	Method
I think they have roughly the same opinion, but there are probably some who think a little differently, so =3	3	O

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening

	Event	Answer	Score	Method
a	Heatwave in the winter season	1 if the heat causes high winds	1	O
b	Heatwave in the summer season	no concern, usually calm weather	0	NA
c	Severe icing (in the winter season)	2 as the boat may capsize and sink	2	O
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	1	1	O
f	Gale	1	1	O
g	Storm	2, perhaps even 3	2.5	O

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

	Event	Answer	Score	Method
a	Heatwave in the winter season	3, [heat wave in winter, often accompanied by storms]	3	O
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	3 (but preferably without the trawl in, then we can get ashore)	3	O
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	0	0	O
g	Storm	3 can even damage the trawl; This is because we have a 40 m boat, but those with 60-80 m boats in the Baltic Sea have no problem turning with trawls during storms in the Baltic Sea	3	O

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Can be repaired on board; 50,000 to repair a trawl, transport it 10,000 SEK; if the entire trawl were destroyed, our trawl costs 200,000 SEK, there are trawls that cost 800,000 SEK, if you lose it Repair costs 50,000-30,000 trawls The boat is rarely damaged; but a 60 m boat got a hole in the side so they had to replace a plate, which cost 20-30,000 SEK, but we go ash	re 10000	50000	sek_u

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
See 11	1e+05	5e+05	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0-2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

	Event	Answer	Score	Method
a	Heatwave in the winter season	2, stays upwind or goes ashore when the wind picks up to storm force, and waits 1-2 days until it calms down; when the waves are too high, the trawl does not fish as well.	2	O
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	2, does not go out	2	O
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	0	0	O
g	Storm	2, does not go out	2	O

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Answer	Score	Method
Depends on the size of the boat, varies quite a lot, 4.	4	O

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

	Event	Answer	Score	Method
a	Heatwave in the winter season	We avoid it, we can go down into the Baltic Sea on the other side of Åland; we can move, we go closer to Finland, 3-4.	3.5	O
b	Heatwave in the summer season	It has no effect	-999	NA
c	Severe icing (in the winter season)	Yes, we can change course if we know in advance, but if the wind is too strong, it doesn't help to move closer to land; we can't go closer than 40 km	4	NA
d	Algal bloom (in the summer season)	No effect	-999	NA
e	Extreme cloudburst (in the summer season)	No effect	-999	NA
f	Gale	No effect	-999	NA

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Answer	Score Method	
Yes, 2.	2	NA

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
We have a few ports where we can land, but there's a small problem at the moment because there have been some ideological issues in Norrsundet, so now we're going to Oxelesund instead. 1	1	NA

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
Yes, I can choose the North Sea or Skagerrak instead if it's too windy in the Baltic Sea. 2	2	O

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score Method	
Not asked	-	NA
	999	

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Answer	Score	Method
I think they think the same (=1 or 2?)	1.5	O

Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

Type	Answer	Score	Method
a stress_injuries	Quite high, it's always there lurking, you're always a little worried, 4, because my 15-year-old son has to go with us.	4	O
b stress_gear	Also stressful	4	NA
c stress_condition	yes, it's also stressful because we lose income, quite high stress = 4	4	O

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Answer	Score	Method
I think it's pretty similar, 2	2	O

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score	Method
no; some have tried to engage in wind power, with smaller boats. =1	1	O

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score	Method
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Yes, we are working on that, herring burgers, producing more products from the Baltic Sea for human consumption; we have to adapt, if it is to be more humane we have to adjust and go ashore after maybe 1-2 days; now we are out for a week; to maintain quality. (=4 or 5?).	4.5	O
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Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1-5; from 1 – very difficult, to 5 – very easy]

Answer	Score	Method
Loans are no problem, we got help developing the equipment, it's much worse today; but you can get help to buy a new trawl =5	5	O

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Answer	Score	Method
large variation between different boats, 5	5	O

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
very important, 5.	5	O

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
we do a little repair work there; we don't land any fish in Fiskebäck; it could be better if we landed there; to a certain extent, but somewhat average, 3	3	O

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1-5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
For my colleagues? Well, you want to have your colleagues, because there are so few of us now, so you want to keep all your colleagues. =5	5	O

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
I think it's very important, when we consume herring, canned herring, it's important, =5	5	O

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Answer	Score	Method
It varies, because we are pelagic, and then you have the demersal fishermen, who are a different species; among the pelagic fishermen, it is quite similar, it is important to everyone.	2	NA

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
It's good to get out on deck, get out on the sea? It's an important part!	4	NA

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Answer	Score	Method
it's strong	5	NA

Q 34 - home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Answer	Score Method	
Yes, Skagen is a really fantastic place, actually very important	5	NA

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Answer	Score Method	
I think everyone thinks pretty much the same	2	NA

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
In the Baltic Sea, from November to June, about half a year, 7 months, at the moment, but that can change; we fish about 80% of those days, are on board and have the trawl in and are fishing	140	170	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
It varies; if we're talking about all boats, we collaborate with another boat, and we've exchanged some fish between us; in the Bothnian Sea, [calculating] 11,000/hour, 25,000 per day in the Bothnian Sea; but off Shetland, we caught 400 tons of mackerel, the entire annual quota, in one hour! That was 8 million/hour	25	30	Million_sek_season

Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit

We are doing extensive renovations now, because we are doing a 10-year class when we go through the entire boat, so it could be 10% of the catch, in a normal year it is 1% of the catch (has an income of SEK 25-30 million per year), so 1% of that. If there are two trawls in a season, SEK 300-400. We have an extra trawl with us, as we have two drums, and we have 1-2 trawls in Norrsundet	1	10	% of income
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Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
if a fishing trip lasts about 1 week; in the Bothnian Sea; 50 m3/day * 6 days* = 240 kkr/week 2.5 tons * 4.5*22 hours; runs 22 hours per day 5000L*8h/day; for one week 240 kkr/week (in the Bothnian Sea) [talks about mackerel fishing]	600	650	liters_hour

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
transport by truck to Skagen from Norrsundet; do not want to drive the boat that far, it costs money; the Bothnian Sea to the Baltic Sea takes 1.5 days	NA	NA	not_reported

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Occasionally, if someone is injured or unavailable; there are four of us on the boat; previously, when we had the net, we needed nine people, but that was ten years ago; substitutes receive the same pay	NA	NA	not_reported

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
No, we don't need to clean the trawls; we lower them into the sea upside down so that they get clean; we both hose them down and dip them; no difference with algae blooms.	0	0	hours

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?
[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Yes, a window can break if you hit a big wave. One boat had a compass on the roof that broke and made a hole right through the roof, so it can cause damage and break things. If water gets into the boat and damages the electronics, it becomes more expensive. 10,000 for a window	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Answer	Score	Method
It varies a lot, many have a very new boat, so the costs are very low, and if you have a larger boat, it can withstand extreme weather better. Astrid loads > 2000 tons, while we load 500 tons, and we have to make 4 trips back and forth, while they only make 1 trip, so they probably have lower fuel costs per fish, even though they consume more when they drive	5	NA

Q 45 - NA

Anything you want to add?
[free text description]

Answer	Score	Method
In terms of extreme weather with storms, I think we've covered everything; this large variation in temperature in the Baltic Sea is something you're looking at, I think, with temperature data from SMHI.	- 999	NA

Interviewee: I 7

7. name: lars ivan hållstrand. category: coastal fisherman id7 is a professional coastal fisherman in the southwestern bothnian sea (öregrund area?). they fish with gillnets and traps. the main fishing season is summer. they can do some ice fishing. the main target is herring, whitefish, salmon, perch. they

have more salmon traps than the average coastal fisherman. fishing for perch (nets) is done from a small boat close to home, while fishing for herring happens with a larger boat more far away. a trip for herring can cost up to 100 litres of diesel (approximately 10 l*hour at 10 knots means that they can go to cast at least 50-60 km from home). how they adapt: they primarily adapt when fishing with nets to hw by changing fishing target. when it's hot they aim for perch and not for herring. in addition, they can perform shorter sets and use cooling boxes. can also adapt to hw and storms by changing fishing location. they are not affected by ab. to avoid hw they can go further out in deeper waters, the opposite for storms. in any case they can move on short distances. their spatial mobility is mainly affected by the boat and infrastructure (they want to go back home) on second instance by management that makes possible to fish in proximity. there is not the same flexibility for salmon traps, which are fixed.

Q 2 - go_out

If the following type of extreme event occurs, how likely is it that you go out and fish?

[1–5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	I probably go fishing anyway, it's the wind that matters, I probably try anyway (=5? or =4?)	4.5	O
b	Heatwave in the summer season	I don't go because the fish won't keep as well, I don't fish for herring at all, in that case I fish for perch, 2	2	O
c	Severe icing (in the winter season)	I am not on the lake at that time of year when there is a risk of icing (because the ice forms so early	-999	NA
d	Algal bloom (in the summer season)	I have not experienced it, but it probably has an impact, so then I don't fish	1	NA
e	Extreme cloudburst (in the summer season)	then I fish anyway	4	NA
f	Gale	depends on the wind direction, if it's N then it's from the outside so then it's difficult, if it's S or SW, in gale force winds I don't fish anyway because there's a strong current in the lake so you get crap in your nets, but I do go out in gale force winds sometimes , depends on the wind direction	3	NA
g	Storm	1	1	O
h	Are there any of these in combination that affect how likely it is that you go out and fish?	storms and weather like that	NA	NA

Multiple options identified

	Event	target	Score (given)
b	Heatwave in the summer season	herring	1

b	Heatwave in the summer season	perch	2
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Q 3 - catchability

Do you experience that the catchability of the fish changes due to such extreme weather events?

[1–5; from 1 – much worse, 3 – no difference, to 5 – much better]

	Event	Answer	Score	Method
a	Heatwave in the winter season	difficult to say, haven't experienced it that much, am more limited when it's windy because seaweed gets caught in the nets (enough if it's gale force), f. fishing gets much worse	-999	NA
b	Heatwave in the summer season	not when it comes to perch, but herring is much more sensitive to heat, so it gets much worse	NA	NA
c	Severe icing (in the winter season)	We don't ask about that	-999	NA
d	Algal bloom (in the summer season)	No, I've only heard that fishing gets worse; I've seen some green filamentous algae on the surface this year (I've seen a little in previous years; it never stays on the surface)	2	NA
e	Extreme cloudburst (in the summer season)	3	3	O
f	Gale	1	1	O
g	Storm	1	1	O

Multiple options identified

	Event	target	Score (given)
b	Heatwave in the summer season	herring	3
b	Heatwave in the summer season	perch	1

Q 4 - catch_assemblage

Do you catch a different composition of species during such extreme weather events?

[1–5; from 1 – very similar, to 5 – very different species composition]

	Event	Answer	Score	Method
a	Heatwave in the winter season	difficult to answer	-999	NA
b	Heatwave in the summer season	herring are fished a little further out, and perch closer in; always fish in different places, so the composition is not affected in each place	2	NA

c	Severe icing (in the winter season)	We skip this one	-999	NA
d	Algal bloom (in the summer season)	Haven't experienced it, but heard that it affects fishing	-999	NA
e	Extreme cloudburst (in the summer season)	1	1	O
f	Gale	Yes, I think it affects the composition; seaweed gets caught in the nets and creates currents; can't say if other species are affected	-998	NA
g	Storm	Yes, I think it affects the composition; seaweed gets caught in the nets and creates currents; can't say if other species are affected	-998	NA

Q 5 - catch_condition

Are the fish in the catch of worse condition or quality when caught during extreme weather events, compared to under normal weather conditions?

[1-5; from 1 – much worse, 3 – no difference, to 5 – much better quality]

	Event	Answer	Score	Method
a	Heatwave in the winter season	3	3	O
b	Heatwave in the summer season	Yes, the warmer it is, the shorter the shelf life of the fish, especially if it is fatty fish (herring, salmon, and whitefish), which is in the summer; herring is particularly sensitive 1	2	M
c	Severe icing (in the winter season)	Not asked	-999	NA
d	Algal bloom (in the summer season)	Cannot answer	-999	NA
e	Extreme cloudburst (in the summer season)	3	3	O
f	Gale	No, I don't think it's the quality, it's probably the catch volume 3	3	O
g	Storm	3	3	O

Multiple options identified

	Event	target	Score (given)
b	Heatwave in the summer season	salmon_whitefish	1.5
b	Heatwave in the summer season	herring	1

Q 6 - cope_condition

If the condition/quality of the fish in the catch is negatively affected, how likely is it that you can do something about it? Explain what!

[1-5; from 1 – very unlikely, to 5 – very likely]

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	Yes, a little (3), I can fish more often, for perch you can keep the nets in for shorter periods, the fish freeze as soon as I get in	3	O
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	-999	NA
g	Storm	NA	-999	NA

Q 7 - lag_effect

Do this extreme weather events have long-term effects on how you fish or catch composition

Of course it has changed, but we don't know why; more storms and such have a certain impact on how I fish, 3-4

	Event	Answer	Score	Method
a	Heatwave in the winter season	NA	-999	NA
b	Heatwave in the summer season	NA	-999	NA
c	Severe icing (in the winter season)	NA	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	NA	-999	NA
f	Gale	NA	3.5	NA
g	Storm	NA	3.5	NA

Q 8 - unc_2_7

How much do you think the answers to these questions (questions 2–7) vary between fishers in the group you represent? 1–5; from 1 – not at all, to 5 – very much

Answer	Score	Method
roughly the same, roughly the same conditions, 2	2	O

Q 9 - personal_safety

How much concern for your personal safety do these extreme weather events bring?

[0–2; 0 – no concern, 1 – personal injuries, 2 – life-threatening]

	Event	Answer	Score	Method
a	Heatwave in the winter season	if the ice becomes unsafe, there is a risk of falling into the water, which can be life-threatening; if you have equipment in the lake, you want to be able to access it, 2	2	O
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	not asked	-999	NA
d	Algal bloom (in the summer season)	0	0	O
e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	2	2	O
g	Storm	2	2	O

Q 10 - damage

How much damage can the extreme weather event cause to your gear and boat? Explain! several answers are possible]

[0–3; 0 – none, 1 – minor damages, 2 – larger damages, 3 – completely destroyed]

	Event	Answer	Score	Method
a	Heatwave in the winter season	0	0	O
b	Heatwave in the summer season	0	0	O
c	Severe icing (in the winter season)	not asked	-999	NA

d	Algal bloom (in the summer season)	Yes, it could be on the gear in the nets, getting clogged, so 1	1	O
e	Extreme cloudburst (in the summer season)	0 , the boat could fill with water, but that would have to be extreme; if you have an open boat, it could happen	1	G
f	Gale	On gear and on the boat; 1	1	O
g	Storm	. 2, gear could be completely destroyed, such as salmon traps and so on	2	O

Q 11 - minor_damage_cost

What are the costs of repairing minor damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
You know about nets, that's minor damage, but salmon traps cost more to repair because they're special equipment; a few thousand for nets, but salmon traps can cost tens of thousands to repair. Yes, on the boat, it's probably if you drive in bad weather, and there is greater damage to the propeller and gearbox on smaller boats, and on large boats it's also the propeller, then it's tens of thousands of kronor.	1000	50000	sek_unit

Q 12 - major_damage_cost

What are the costs of repairing major damages to gear and boat?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
see 11	50000	100000	sek_unit

Q 13 - change_practices

Do you change anything in how you normally fish when such an extreme weather event occurs?

[0-2; 0 – change nothing, 1 – change a little, 2 – change a lot + free text description]

	Event	Answer	Score	Method
a	Heatwave in the winter season	Yes, if there is no ice, I cannot fish, so I have to fish from a boat, but that is usually not very productive; 2	2	O
b	Heatwave in the summer season	1 changes how much perch I fish compared to herring	1	O

c	Severe icing (in the winter season)	I don't know, I haven't experienced it (maybe I'll skip it then),	-999	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	0	0	O
f	Gale	Changes the fishing spot	1	NA
g	Storm	2; Do you go out in a storm? It has happened that you make a mistake and go out, but you have to put the nets in a sheltered spot, you have your spots	2	O

Q 14 - unc_9_13

How much do you think the answers to these questions (questions 9–13) vary between fishers in the group you represent?

Answer	Score	Method
They probably have similar conditions, so they probably answer quite similarly; =2	2	O

Q 15 - avoidance

How easy is it to avoid an extreme weather event by going to another fishing location?

[1–5; from 1 – very difficult to avoid, to 5 – very easy to avoid]

	Event	Answer	Score	Method
a	Heatwave in the winter season	1 (if the ice is not strong enough, it's pretty much the same, then you have to go very far),	1	O
b	Heatwave in the summer season	I don't know, but you can still fish for perch; the roach moves further out, and I can go a little further out, but I don't have a very big boat, so I can't go that far out, 2	2	O
c	Severe icing (in the winter season)	I think so , it depends on how it accumulates in certain places	4	NA
d	Algal bloom (in the summer season)	NA	-999	NA
e	Extreme cloudburst (in the summer season)	1 (but it doesn't matter)	1	O
f	Gale	4, depends on the wind direction, and sometimes the fishing spots don't match where it's sheltered.	4	O
g	Storm	same there, but you rarely fish in a storm if you know about it, but sometimes you have to go out as I said	4	NA

Q 16 - gear characteristics

Do you have gear, equipment and boat that allow you to choose other fishing locations (to avoid an extreme weather event)?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, within short distances, 2 – yes, also at long distances]

Answer	Score (self)	Score (given)
I have mobile gear (perch can move distance =1), herring =1 (somewhat limited by boat), but I cannot choose the location for fixed gear (= 0)	M	NA

Multiple options identified

target	Score (given)
not_specified	1
salmon	0

Q 17 - infrastructure

Is there infrastructure on land (e.g., landing harbors, boat ramps) that makes it possible to choose other fishing locations?

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score	Method
Extremely, there are other places to go ashore; fixed with the salmon because then you have to report the landing port before you unload them (and when you unload); for the herring, I have the shaking machine in the boathouse, so it is difficult to find out the catch elsewhere, but I can go to an emergency port, i.e. =0	0	O

Q 18 - management_space

Do current management regulations make it possible to choose other fishing locations? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score	Method
yes, in a limited area; the license determines where you can fish; we have quotas and fixed gear in fixed locations for salmon, but with mobile gear you can move around in the local area at least; =1	1	O

Q 19 - management_species

Do current management regulations make it possible to choose other target species? Briefly describe what possibly limits your choice.

[0–2; 0 – no, requires that I am at the same place, 1 – yes, to some extent, 2 – yes, much that makes great mobility possible]

Answer	Score	Method
There isn't much else to fish for; we catch other fish species, but in much smaller quantities, such as pike and pikeperch; salmon is subject to quotas; herring too, but we don't fill the coastal quota, and haven't done so for several years.	999	NA

Q 20 - unc_15_19

How much do you think the answers to these questions (questions 15–19) vary between fishers in the group you represent?

Answer	Score	Method
Not everyone has salmon traps, but otherwise it's pretty similar; =2.	2	G

Q 21 - stress

How much stress does the risk of the following events cause (in general)?

[1–5; from 1 – really not stressful, to 5 – very stressful]

Type	Answer	Score	Method
a stress_injuries	1 no, I don't think about it much	1	O
b stress_gear	sometimes, when the weather is bad, windy, stormy, that's when you feel like you have stuff out there, it's mostly the salmon traps you think about; you usually have time to remove the movable gear if you know bad weather is coming; 4	4	O
c stress_condition	I never think about it, 1	1	O

Q 22 - unc_21

How much do you think the answers to this question about stress (question 21) vary between fishers in the group you represent

Answer	Score	Method
I would think they would answer the same, =1	1	O

Q 23 - job_mobility

If your income from fishing decreases, can you complement or replace it with other, non-fishing related, activities to gain income

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score Method	
a little; 3	3	O

Q 24 - entrepreneurship

How likely is it that you explore new techniques or strategies for fishing and marketing of your fishing products

[1–5; from 1 – very unlikely, to 5 – very likely]

Answer	Score Method	
I try to develop fishing so that it is as easy as possible to do certain things, and I try a little bit to find new ways to sell. =4	4	O

Q 25 - credit

How easy is it to gain access to credits, financing or loans for investments

[1–5; from 1 – very difficult, to 5 – very easy]

Answer	Score Method	
It probably depends on how much money and what for. You need to have good ideas. So, to buy a salmon trap? It's a bit uncertain because salmon stocks are low at the moment. A filleting machine? It's probably not entirely impossible.	2	NA

Q 26 - unc_23_25

How much do you think the answers to these questions about economic flexibility (questions 23–25) vary between fishers in the group you represent

Answer	Score Method	
I think it would vary a bit; some don't do much processing and mostly want to fish, so 4	4	O

Q 27 - knowledge_transmission

How important is it to you to pass on the profession, the tradition, the knowledge and/or the leisure activity in the form of fishing to future generations

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score Method	
I've given up. I wish it could continue, but it's so difficult, there are so many problems, with the authorities, among other things. Yes, I think it's important. I'm interested, but the financial aspect is difficult for a new start. =5	5	O

Q 28 - benefit_local_community

How much does your local community benefit from your fishing

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
They get fresh fish nearby; quite a few people shop there, people come to his home to buy fish, and not just when he's at the market; =5	5	O

Q 29 - benefit_personal_community

How much does your “community” benefit from your fishing? Explain!

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
No, there are so few who fish. Do you share experiences? Yes, we benefit from each other, in the past they couldn't talk to each other. Sometimes it was a matter of life and death, you've heard about that, it was when there were several in each village, but now there are so few of us that it's easy to get along. =4	4	O

Q 30 - traditional_food

How important is your fishing for providing traditional food

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
5, that's what I do.	5	O

Q 31 - unc_26_30

How much do you think the answers to the questions about the contribution of fishing to society (questions 27–30) vary between fishers in the group you represent

Answer	Score	Method
Some, and some would say differently, =3	3	O

Q 32 - outdoor_benefit

How important is it for you to be outdoors (when you fish) for your well-being

[1–5; from 1 – not important at all, to 5 – very important]

Answer	Score	Method
yes, it is quite important, it is very important =5	5	O

Q 33 - identity

How much does fishing contribute to your identity and self-image

[1–5; from 1 – very little, to 5 – very much]

Answer	Score	Method
quite a lot (=4 or 5? [He doesn't use any superlatives!])	4.5	O

Q 34 - home

How much does fishing contribute to creating a bond to local places

[1–5; from 1 – very little, to 5 – very much]

Answer	Score	Method
yes, but I usually go to the same place, and stupidly enough, I don't try out many different places; some places are special to visit =4?	4	O

Q 35 - unc_31_34

How much do you think the answers to the questions about personal well-being (questions 32–34) vary between fishers in the group you represent

Answer	Score	Method
They would probably be similar; =2.	2	O

Q 36 - fdays

Approximately how many days per year do you fish?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
around 200 days	NA	200	days

Q 37 - income

Approximately how large is your gross income from a fishing day on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit

It depends on processing, fishing, and smoking; it varies at different times of the year. Spring and early summer are quite intense, then it slows down. You don't get your money back for surströmming until you've sold the cans. Varies from SEK 10,000-50,000/day, depending on the catch and salmon and such; then it is processed. If you get 100 salmon in a trap in one day, that's worth quite a lot just raw, then you smoke it and save it, so you can spread it out over the	year 1000	50000	se
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Q 38 - season_maint

How large are the repair and maintenance costs for the fishing during one year?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Nets require quite a lot of work, and salmon traps in the spring; say SEK 10,000-35,000/year. And in terms of time? About 30 days, depending on whether you encounter breakdowns and such.	10000	35000	sek_season

Q 39 - fuel_trip

How large are the fuel costs (or fuel consumption) for a fishing trip on average?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
30-50 liters of diesel for herring; if fishing for perch, the costs are not so high because it is quite close, then 4-5 liters of gasoline (because it is a small boat).	5	50	liter_trip

Q 40 - extra_fuel

What are the costs for extra fuel to travel further than normal?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
100L of diesel for herring	50	100	liter_trip

Q 41 - extra_crew

What are the costs for extra crew on board?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
has one specifically for herring fishing; he does not fish that many days; costs SEK 60,000-100,000/year	60000	100000	sek_year

Q 42 - extra_cleaning_cost

What are the costs for gear cleaning (in case of unusual fouling, e.g., due to algal bloom)?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
Rinses the salmon traps, takes a few hours in both spring and fall, approx. 20 hours; extra work if there is algae blooms; sometimes there is growth and sometimes there is moss floating around, which is difficult to rinse off, so it has to lie there and rot first; it grows on what is on the beaches and if it is windy, it comes loose and gets stuck in the trap	20	NA	hours

Q 43 - exytra_other

Are there any other extra costs that may arise during extreme weather events, and how large are they then?

[free text description]

Answer	Score - low (given)	Score - up (given)	Unit
No, it's usually repairs if there is any damage, etc	NA	NA	not_reported

Q 44 - unc_36_42

How much do you think the answers to these questions (questions 36–43) vary between fishers in the group you represent?

Answer	Score	Method
Yes, I think so, 5	5	O

Q 45 - NA

Anything you want to add?

[free text description]

Answer	Score	Method
I don't have much experience with it yet, but what we've seen is milder winters and more wind; things have changed, but we don't know what the cause is.	- 999	NA